

Market Fragmentation and Inefficiencies in Maritime Shipping

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Abstract

Problem definition: Many large-scale transportation systems rely on decision-makers who independently allocate resources across space and time. While decentralization fosters competition, it can also generate substantial operational inefficiencies. We study this trade-off in the global oil tanker market, where approximately half of all miles are traveled without cargo (ballasting). Although some empty repositioning is unavoidable due to geographic imbalances in oil trade, the extent to which market structure contributes to this inefficiency has not been quantified. *Methodology/results:* We develop a framework that decomposes observed empty miles into shares attributable to trade imbalances, demand uncertainty, and market fragmentation. Using granular commercial data, we show that fragmented ownership accounts for 7–16% of total empty miles, a “fragmentation tax” roughly twice the share attributable to demand uncertainty. These inefficiencies arise because independent owners optimize over limited fleet footprints and cannot internalize cross-fleet balancing externalities. Our decomposition also isolates the share of emissions induced by ownership fragmentation from that caused by structural trade patterns, identifying the portion of carbon emissions that is operationally addressable. Focusing on the growing industry trend of shipping pools, we show that even modest consolidation captures most of the gains achievable under full centralization, while preserving competitive market structure. *Managerial implications:* Our findings provide quantitative evidence that modest consolidation in large-scale spatial allocation systems can deliver substantial efficiency gains without requiring high market concentration, establishing operational coordination as a practical complement to technological and regulatory approaches that currently dominate the decarbonization agenda.

Keywords: Transportation networks, Decentralization, Market fragmentation, Resource pooling, Ballasting, Supply chain sustainability.

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1 Introduction

Many transportation and logistics systems allocate resources across space and time under decentralized control. While fragmentation preserves competition and limits market power, it can generate significant inefficiencies when repositioning decisions are made independently. Quantifying the operational cost of decentralization, and distinguishing what is unavoidable from what is induced by the market structure, is central to understanding efficiency in networked markets and to designing effective policy responses.

We study this question in the global oil tanker industry, a setting that combines large geographic imbalances, substantial repositioning, and high fragmentation. Maritime transportation carries over 80% of international trade by volume, and the oil tanker segment alone accounts for roughly 30% of seaborne trade and approximately one-quarter of all shipping emissions (UNCTAD, 2025). A defining operational feature of this market is that nearly half of all miles traveled are empty, a practice known as ballasting. Because oil production is geographically concentrated while consumption is dispersed, some empty repositioning is unavoidable. However, the market is also highly fragmented, with no single ship operator controlling more than 5% of global tanker capacity. This raises a fundamental question: how much of the observed inefficiency is dictated by geography and uncertainty, and how much is a consequence of decentralized decision-making?

This distinction has immediate policy relevance. The urgency of decarbonizing maritime shipping, which contributes nearly 3% of all greenhouse gas emissions (International Maritime Organization, 2020), has prompted a range of regulatory and technological responses. The International Maritime Organization approved a Net-Zero Framework in April 2025 that includes mandatory fuel intensity standards and a global carbon pricing mechanism, and the European Union incorporated shipping into its Emissions Trading System in January 2024. On the technology side, alternative fuels such as green ammonia and hydrogen are widely regarded as the most promising long-term solutions, but they require cumulative investments estimated at \$1 to \$1.4 trillion through 2050 (Krantz et al., 2020), and fleet turnover cycles of 20 to 25 years impose a natural constraint on adoption speed. Critically, existing assessments suggest that price-based instruments alone may yield only modest near-term reductions: Pereda et al. (2025) estimates that a \$100 per metric ton of CO₂ carbon tax on international shipping would decrease emissions by only 3.6%. Given these constraints, understanding whether market structure constitutes an independent and actionable lever for efficiency improvement is of primary importance.

In this paper, we provide a quantitative decomposition of the sources of ballasting inefficiency. We distinguish among three drivers. The first is trade imbalances, which determine the empty miles that can be attributed to the geography of oil supply and demand. The second is demand

uncertainty, arising from the stochastic timing and location of future load requests. The third is market fragmentation, the additional inefficiency generated by decentralized decision-making across many independent operators. We refer to this last component as the “*fragmentation tax*”: the share of empty miles and associated emissions attributable specifically to the structure and operations of the oil tanker industry, beyond what geography and uncertainty require. We choose this term to emphasize that fragmentation imposes a cost, both financial and environmental, that is structural in nature, and that policy instruments failing to account for this distinction risk penalizing carriers for factors outside their control while leaving the most actionable source of inefficiency unaddressed.

Our analysis leverages a unique data set that covers the voyages of about 6,000 oil tankers between 2018 and 2022, obtained through a partnership with a leading tanker market analysis provider. Unlike studies relying solely on Automated Identification System (AIS) signals, our data include the identity of the commercial operator responsible for each vessel and voyage. This allows us to link load and positioning decisions to the entities that make them and to study how the induced decision-making structure affects market outcomes. We combine these data with integer programming and stochastic dynamic optimization models to compute optimal vessel-to-load assignments under alternative informational and organizational regimes, holding realized laden demand fixed.

The decomposition yields results that are both clarifying and, in some respects, striking. Trade imbalances account for the majority of empty miles, explaining 73-77% for smaller vessel classes, and up to 88% for Very Large Crude Carriers (VLCCs). This is expected and consistent with the geographic structure of global oil flows. What is less anticipated is the magnitude of the fragmentation component. Market fragmentation accounts for 7-16% of total empty miles, depending on vessel class, a share that is economically significant and roughly double the contribution of demand uncertainty. This suggests that a substantial portion of avoidable inefficiency in this market is attributable to the way the industry is organized. Markets served by smaller tankers, e.g., the Aframax class, which is the fastest-growing segment, are especially susceptible, both because fragmentation is higher due to lower barriers to entry and because the larger set of feasible routes makes coordination more complex. For Aframax vessels, fragmentation alone explains roughly 7.5% of all carbon emissions, ballasting and laden combined. To put this in perspective, this exceeds by a factor of two the estimated emission reductions from a \$100 per metric ton of CO₂ carbon tax on the entire shipping industry, currently the most prominent policy instrument under consideration.

These optimization-based estimates find direct support in the data. Using an instrumental-variables approach that exploits variation in vessel ownership concentration across geographic markets and time, we show that markets with higher concentration exhibit significantly shorter ballast distances and lower emissions. Moving from the 25th to the 75th percentile of market concentration

is associated with a roughly 25% reduction in ballast emissions. The convergence of two methodologically distinct approaches, i.e., one based on simulations and one on observed market variation, strengthens our confidence that fragmentation is a first-order driver of inefficiency in this market.

These findings naturally lead us to investigate whether practical interventions can address fragmentation. We focus on shipping pools, joint ventures in which independent shipowners place vessels under unified commercial management while retaining ownership. Pools are already a growing feature of the industry: Tankers International operates the world’s largest VLCC pool with approximately 65 vessels, Hafnia manages around 200 product tankers across six specialized pools, and pooled tankers account for roughly 17% of the total fleet by capacity. Because pools remain small relative to the overall market, each controlling no more than 5% of fleet capacity, they do not confer meaningful market power. Antitrust regulators have historically taken a more permissive view of pooling arrangements than of formal mergers or shipping alliances, precisely for this reason.

Our simulations reveal that consolidating vessels into modest-sized pools of approximately 60 vessels, e.g., corresponding to roughly 5% of the global Aframax fleet, captures 80-90% of the efficiency gains achievable under full centralization. At the operator level, we show that the gains arise through two reinforcing channels. Larger fleets experience shorter ballast delays, i.e., the excess time beyond what direct transit would require, through greater scheduling flexibility and a larger set of vessels across positions, enabling faster matching between available capacity and cargo opportunities. They also operate broader port networks that support route diversification and reduce dependence on individual trade lanes, enabling the fleet-level matching improvements that underlie the market-level gains documented above. In aggregate, these coordination gains can reduce ballasting-related emissions by up to 15%.

Taken together, our findings identify operational coordination as a practical, near-term complement to the technological and regulatory strategies that dominate the decarbonization agenda for maritime shipping. By quantifying, for the first time, the relative contributions of trade imbalances, uncertainty, and fragmentation in a large-scale transportation market, we establish in a data-driven fashion both the magnitude of the opportunity and the modest level of consolidation required to capture it. More broadly, the framework and findings may generalize to other settings in which independent operators make repositioning decisions, such as container shipping, trucking, and ride-hailing, suggesting that ownership structure and the coordination losses it induces deserve greater attention as determinants of efficiency in networked transportation markets.

1.1 Related Literature

This paper studies how market fragmentation drives operational inefficiencies in the global oil tanker market and contributes to several strands of literature spanning transportation coordination,

maritime economics, resource pooling, and operational sustainability.

A central theme of our paper is that fragmented ownership of transportation assets generates excessive empty repositioning. Coordination mechanisms aimed at reducing empty repositioning have been studied across a variety of transportation settings. For example, in trucking [Ergun et al. \(2007\)](#) show that shipper collaboration can substantially reduce empty truck miles by constructing continuous-move tours; in liner shipping, [Agarwal and Ergun \(2010\)](#) design optimal network and capacity allocation schemes to sustain container carrier alliances. At a higher level, [Séjourné et al. \(2018\)](#) analyze a decentralized transportation market and show that fragmentation reduces efficiency because small fleets struggle to reposition assets when demand is randomly split among them. Although these papers demonstrate theoretically that coordination can improve efficiency, little is known about the magnitude of these gains in practice. We address this gap by quantifying, using granular commercial data from a global market, the magnitude of efficiency losses attributable to fragmentation and the share recoverable through realistic coordination mechanisms, i.e., shipping pools. In a different operational setting, [van den Berg et al. \(2025\)](#) pursue a related empirical agenda by quantifying the value of spatial coordination for ambulance fleets.

Our analysis distinguishes between two conceptually distinct sources of inefficiency in spatial systems: those driven by demand (geographical imbalances and stochastic arrivals) and those driven by market structure (fragmented ownership and limited network coverage). A large literature examines pricing (e.g., [Cachon et al. 2017](#); [Bimpikis et al. 2019](#); [Besbes et al. 2021](#); [Ma et al. 2022](#)) and relocation policies (e.g., [Braverman et al. 2019](#); [Özkan and Ward 2020](#); [Hosseini et al. 2025](#)) in centrally operated ride-sharing systems, characterizing optimal responses to demand imbalances under unified control. In contrast, we study a fragmented market in which independent operators control subsets of assets and cannot internalize system-wide positioning externalities. While our analysis utilizes centralized optimization problems, we use them as benchmarks to measure the inefficiencies induced by decentralized ownership rather than to prescribe optimal policies.

Within maritime shipping, several recent papers analyze inefficiencies arising from search frictions and contractual flexibility. [Prochazka et al. \(2019\)](#) and [Adland and Prochazka \(2021\)](#) estimate the value of foresight and contracting flexibility in dry-bulk markets, focusing on individual operators and relying on simulated data calibrated to aggregate statistics. [Brancaccio et al. \(2020\)](#) and [Brancaccio et al. \(2023b\)](#) exploit voyage-level data to structurally estimate search frictions and pricing inefficiencies. Their focus is on how imperfect matching distorts equilibrium prices and relocation decisions within a given market structure. In contrast, we examine how the allocation of vessels across independent owners shapes decision-making and efficiency at the market level, given that each owner optimizes over a limited geographic footprint and cannot internalize cross-fleet balancing externalities. Because empty miles arise even in the absence of search frictions, owner-

ship fragmentation as a structural source of inefficiency has remained underexplored. We provide evidence that ownership structure exerts a first-order effect on ballasting inefficiencies and quantify its magnitude in maritime shipping.

Recent research has examined matching quality between supply and demand in transportation systems with spatial and temporal imbalances (e.g., [Azagirre et al. 2024](#); [Castro et al. 2023](#)). Although the optimization problems we solve can be recast as dynamic matching problems, our objective is not to derive approximately optimal policies but to quantify the inefficiencies induced by decentralized market structure.

Our examination of shipping pools builds on the literature on resource pooling in operations management. Pooling has long been studied in inventory systems ([Eppen, 1979](#); [Bimpikis and Markakis, 2016](#)), manufacturing flexibility ([Jordan and Graves, 1995](#); [Netessine et al., 2002](#); [Liu et al., 2016](#)), and transportation alliances ([Wright et al., 2010](#)). More recently, [Jacquillat et al. \(2024\)](#) show that capturing the full value of pooling may require combining complementary sharing dimensions. We extend this literature by quantifying, at a market-wide level using operational data, the efficiency gains attainable through partial consolidation. Our results show that relatively small pools capture most of the gains from full centralization, highlighting the nonlinear returns to coordination in spatial systems.

Our finding that operational coordination can meaningfully reduce ballasting-related emissions situates our work within the growing literature on carbon emissions in supply chains and transportation. Prior research emphasizes the importance of accurate emissions attribution and the role of allocation rules in shaping environmental incentives ([Caro et al., 2013](#); [Sunar and Plambeck, 2016](#)). By developing a decomposition framework tailored to maritime shipping, we distinguish emissions driven by structural trade imbalances from those arising endogenously from market fragmentation. This separation identifies the portion of emissions that is operationally addressable without technological change. More broadly, while maritime policy discussions largely focus on technological innovation and regulatory instruments such as Emission Control Areas ([Zhuge et al., 2024](#)), we identify market structure itself as a primary driver of emissions. This perspective complements evidence that organizational practices and design choices from electrification to green certification materially affect environmental outcomes ([Delmas and Toffel, 2008](#); [Arora et al., 2024](#); [Feldman et al., 2025](#); [Fatehi and Honhon, 2025](#)). Ultimately, while micro-level optimizations like speed and timing can reduce emissions at the individual route level ([Franceschetti et al., 2013](#)), our work establishes market structure as a critical macro-level lever for sustainability that requires no technological shift. Thus, we emphasize how modifying market conditions can incentivize agents to make decisions that are mutually beneficial to them and the environment, similarly to the recommendations to improve retail EV adoption ([Kaps et al., 2025](#)).

2 Oil Shipping: Context and Data

Shipping activity can be broadly divided into three categories. Raw materials (dry bulk, oil, and other liquid products) collectively constitute 85% of global volume; containerized cargo accounts for approximately 11%; and the remaining portion consists of specialized categories such as chemicals.¹ The oil transportation sector alone accounts for 30% of seaborne trade volume, making it one of the most significant markets in maritime transportation. While extensive land infrastructure exists to move crude oil from extraction sites to refineries, an estimated 61% of the daily production volume of 90 million barrels relies on maritime transport.^{2,3} A fleet of about 8,800 ocean-going oil tankers operates in this industry, divided into six independent segments based on cargo capacity, called *vessel classes* (see Table 1). The smallest vessels have a capacity of about 300,000 barrels of oil, while the largest can accommodate nearly ten times that volume. Ships within each class are relatively homogeneous in size, so that they can access the same ports and use the same waterways.

Vessel class	Min DWT	Max DWT	No. vessels
MR1	25,000	42,000	937
MR2	42,000	60,000	1,800
Panamax	60,000	80,000	475
Aframax	80,000	125,000	1,146
Suezmax	125,000	200,000	661
VLCC	200,000	-	867

Table 1: Deadweight-tonnage (DWT) capacity and number of ships in each vessel class in our data.

While major oil producers and refiners maintain modest fleets of tankers, the bulk of the industry relies on a network of small shipowners who make their vessels available for leasing. Hiring of oil tankers occurs through a variety of contract types, characterized by two dimensions: the length of time for which a vessel is hired, and the financial obligations of the parties involved. Vessels may be chartered for individual journeys between a load port and a discharge port, or for longer periods of time (e.g., a year). In return, the hiring party pays a daily rate to the owner for the duration of the contract. Vessels are predominantly engaged on a per-voyage basis, on what is often referred to as a *spot* market. Even vessels hired for longer periods are often subleased to other exporters to take advantage of rate fluctuations, so that virtually all loads are transported under spot contracts.

The spot market for maritime shipping operates through a network of brokers who facilitate transactions between oil companies (exporters) and carriers. A typical transaction begins when

¹https://unctad.org/system/files/official-document/rmt2022_en.pdf

²<https://www.bp.com/content/dam/bp/business-sites/en/global/corporate/pdfs/energy-economics/statistical-review/bp-stats-review-2022-full-report.pdf>

³https://www.eia.gov/international/analysis/special-topics/World_Oil_Transit_Chokepoints

an exporter needs transportation and contacts a broker, providing essential information about the cargo: type, loading and unloading locations, and the desired departure schedule. The broker then presents these cargo requirements to select vessel operators whose ships are positioned near the departure area, and works to finalize an agreement between the parties. The resulting contract (known as a *fixture*) specifies the exact location and day of loading, along with the daily vessel rental rate. This rate is paid only for the duration of the trip from the loading port to the final destination. Because the timing and duration of sea voyages are inherently uncertain, this process typically begins shortly before the intended loading time. Consequently, only vessels located close to the loading area can realistically service a given cargo. For this reason, operators’ decisions about where to position their vessels are critical to securing profitable contracts, as strategic positioning directly impacts their ability to compete for available cargoes.

To see why positioning matters, consider a ship operator’s perspective right after discharge. It is unlikely that the vessel has already been contracted for a new load.⁴ The operator must decide whether to wait for new offers in the same port (or neighboring anchorages), or to relocate somewhere else. During this phase, the shipowner typically receives proposals from brokers for new cargoes whose load area is a few days away from the vessel’s current position. If the offer is fixed, a new *voyage* begins, which ends when the new discharge port is reached. Each voyage can thus be divided into an initial *ballast* leg, where the tanker is empty, and a final *laden* leg, where the tanker is full.

The market dynamics described above naturally lead to *empty miles*, a feature common to virtually all transportation markets. Empty miles substantially affect the operating costs of oil shipping companies and impose a significant burden on the environment. Because of the economic and environmental costs associated with ballasting, understanding its causes and exploring potential remedies are of primary importance.

2.1 Data Description

We partnered with a shipping and analytics company, which provided us with a list of 234,795 voyages completed from January 1st, 2018 to March 26th, 2022 by 5,886 vessels; [Table 1](#) offers a breakdown by vessel class. For each voyage we have the identifier of the vessel, as well as its load and discharge ports with the respective arrival and departure dates; we also observe a *starting port* that corresponds to the discharge location of the previously completed voyage. The ballast leg of the voyage is the trip from the starting port to the load port, and the laden leg goes from the load port to the discharge port. We therefore observe the duration of each voyage leg and can recover

⁴Exporters are reluctant to charter a tanker that has not yet discharged its previous cargo, because of potential delays associated with port congestion, discharging, and refurbishing of vessels.

how much time each vessel spent docked, ballasting, or sailing laden.

Geographical information is aggregated at various levels, from port level (the finest partition) to *wide areas* (the coarsest). We conduct our analysis at different granularities, since the most appropriate level depends on the specific application. Generally, we employ port-level, *narrow area*, and *intermediate area* information: narrow areas partition the world into 51 regions, while intermediate areas partition it into 28. A list of the intermediate areas can be found in [Table 16](#).

In addition to geographical data, we have information about fuel consumption, pollutant emissions, speed, and the name of the *Commercial Operator* that managed the vessel for that voyage. The commercial operator is the agent responsible for deciding which loads a vessel carries and where it ballasts. The commercial operator may not correspond to the actual shipowner, since vessels with different owners may be part of the same shipping pool, operated by a single entity.⁵ We do not have access to contracted rates; we discuss how this limits our analysis in [Section 3.3](#).

Our dataset includes only completed voyages; we do not have information on potential demand that was not fulfilled. However, several features of this market imply that potential demand closely aligns with realized loads. Demand for oil transportation exhibits generally low price elasticity, because shipping costs represent a small fraction of an oil producer’s total profit, making exporters unlikely to forgo shipping due to price fluctuations in transportation rates. Furthermore, the high capital costs associated with terminal storage and the absence of viable transoceanic alternatives once oil reaches a terminal create a “no outside option” environment. Given that spot contracts are typically finalized on short notice when loads are already prepared for departure, the likelihood of an exporter engaging a broker and subsequently forgoing the contract is minimal, so observed laden trips reliably reflect the underlying distribution of demand for transportation.

Finally, we do not consider voyages for which we do not have information about the commercial operator (15% of voyages). Moreover, we restrict the analysis to vessels for which we have a complete history, i.e., those whose position can be accounted for from the first time they are assigned a load to the last time they discharge. This does not rule out vessels that undergo maintenance or become unavailable for other reasons, which we explicitly account for in [Section 3](#), as long as we can verify that we did not miss any loads they fulfilled. This leaves 5,658 tankers and 194,455 voyages (approximately 82% of the dataset); we now summarize the salient features of the oil shipping market as they emerge from our data.

Trade imbalances. Oil reserves are unevenly distributed across the globe, leading to substantial imbalances in oil transportation patterns. These imbalances are reflected in our dataset, as shown in [Table 16](#) and [Figure 3](#), and represent the primary driver of ballasting. For instance, the

⁵The same ships may, and in fact do, operate under different pools throughout the time span of our dataset.

“China/Taiwan” region witnessed a threefold surplus of cargo arrivals over departures. Conversely, in regions such as the “Arabian Gulf” (i.e., Persian Gulf), there was a nearly five-fold surplus of cargo departures relative to arrivals. Given these patterns, vessels rarely find a load to transport at the same location where they discharge.

Ballasting. Ballasting is a pervasive feature of the voyages we observe. Approximately 95% of voyages begin in a port different from the one where the vessel had just discharged, and in 55% of cases the tanker must change geographical area to reach the new load port. On average, 40% of the length of each voyage consists of the ballast leg, with minor differences across vessel classes. This translates to an annual consumption of approximately 10 million tons of fuel for ballasting alone, equivalent to 32 million tons of CO₂ emissions (comparable to the annual output of 7 million passenger cars) at an estimated cost of 7.2 billion USD (September 2023 prices). [Table 14](#) summarizes this information. Ballasting typically occurs because vessel operators judge the likelihood of finding new loads in their current location to be sufficiently low that paying the cost of relocating is justified.

Market fragmentation and pools. A distinctive feature of our dataset is that it includes information about the commercial operators of the tankers, allowing us to construct a detailed picture of market concentration, its evolution over time, and how fleet size correlates with operator decisions. The control structure is extremely fragmented. As evidenced in the left panel of [Figure 4](#), a substantial number of operators have fleet sizes ranging from 1 to 3 vessels, while very few control more than 10. The fleet size of the largest commercial operators exhibits notable variation over time, as shown in [Figure 5](#), while smaller operators tend to maintain more stable fleets. This difference can be attributed to the fact that large commercial operators manage vessels via *shipping pools*, adding to their own fleet vessels owned by others (typically small shipowners themselves). A shipping pool is formed when a number of shipowners agree to place their vessels under a single manager. The pool manager assumes responsibility for all vessels and maximizes collective revenues, while the respective owners remain entitled to a share of the earnings the pool generates.

Commercial operators can use shipping pools as a more flexible tool for expanding their fleets compared to formal mergers or acquisitions.⁶ Pools have become increasingly popular thanks to advances in computational capabilities that enable real-time optimization. Despite their advantages, shipping pools’ market share remains low. [Table 17](#) shows that for the Aframax vessel class the largest operator controls less than 4% of the total number of vessels, with the top 20 operators

⁶Vessels can ask to join a pool or withdraw at any time, although either process may take between one and six months before being finalized.

collectively controlling less than 40% of the asset share.⁷ Similar patterns prevail across other vessel classes, with a marginal uptick in concentration for larger tankers.

3 Quantifying the Fragmentation Tax

The description of the oil shipping market in [Section 2](#) points to three main sources of ballasting inefficiency: trade imbalances, uncertainty about the time and location of future loads, and market fragmentation. The role of trade imbalances is intuitive: too few loads originate from some of the most common destinations, forcing tankers to sail away empty. Uncertainty generates ballasting because tankers may depart a port even when a load would have been offered shortly after; while obvious in hindsight, the operator assessed that the probability of finding a profitable load was too low to justify waiting. Finally, market fragmentation contributes to ballasting through several channels. Commercial operators often have access to different information due to their diverse broker networks, creating information asymmetries. Operators managing larger fleets have greater flexibility in positioning their vessels, allowing them to be more opportunistic in pursuing profitable contracts. Rather than attempting to isolate each channel individually, we group them under the term *market fragmentation*, which captures their common underlying cause: had the market been more consolidated, many of these asymmetries would have been eliminated.

The goal of this section is to quantify the share of ballasting attributable solely to market fragmentation. This serves as an upper bound on the efficiency gains achievable by any market-based intervention that does not alter the underlying pattern of oil flows. Along the way, we also measure the extent to which ballasting can be attributed to the other two sources. We first quantify the share of empty miles due to trade imbalances as a baseline level of ballasting intrinsic to oil trade patterns. Intuitively, this baseline is defined by the minimal amount of empty miles that the transportation system must incur to serve all demand when loads are known in advance and vessels are controlled by a single entity. The share of ballasting due to fragmentation is then the *additional* amount above this baseline that arises when decision-making is split among all the different commercial operators.

Assumptions. To perform the analysis of this section, we use data from all four years of voyages in our dataset to estimate emissions and travel times between geographical areas. Given that the COVID-19 pandemic disrupted oil trade flows, we then restrict our focus to the pre-pandemic period, from January 2018 through December 2019.⁸ We discretize time into five-day periods, so

⁷By comparison, the container shipping market is far more concentrated: the four largest companies control 60% of capacity.

⁸This also ensures consistency with the analysis of [Appendix A](#), which is based on reinforcement learning using synthetic data calibrated to pre-pandemic market trends.

that, for example, voyages departing from January 1st, 2018 to January 5th, 2018 are assigned to period $t = 0$; consistently, for this analysis only we drop voyages taking less than 5 days from start date to discharge date. We use the intermediate geographical areas described in [Table 16](#). We impose that if vessels reach a location at time t , they are also able to load cargoes at time t .⁹ Finally, we assume that (i) the laden leg of each voyage has the same duration irrespective of the ship assigned to it; (ii) loads are fungible within a vessel class, so that each can be transported by any available vessel in that class, but not across classes.¹⁰

3.1 Trade Imbalances

We define the share of ballasting due to oil trade imbalances as the minimal amount of ballasting required to balance the transportation system. In our approach, we take demand as an exogenous input and focus on the transportation plans that satisfy all load requests observed in the data. As discussed in [Section 2.1](#) and [Section 3.3](#), demand for oil transportation is inelastic with respect to freight rates, and therefore laden trips provide a reliable measure of underlying demand (at least in the short/medium-term), irrespective of the market structure. The share of ballasting exclusively attributable to trade imbalances can then be estimated as the cost that would obtain if transportation were organized by a clairvoyant entity that controls all vessels and minimizes empty miles while satisfying all observed demand, expressed as a fraction of the total cost of empty travels measured in the dataset. Because this entity optimizes in hindsight, uncertainty plays no role; and because it controls all vessels, there is no loss due to fragmentation. Since the different vessel classes define virtually independent markets, we optimize separately within each class.

Formally, we cast each problem as an integer linear program whose objective is to minimize total ballasting cost defined in terms of CO₂ emissions. Since we impose the constraint that all observed voyages must be fulfilled, minimizing ballasting costs is equivalent to finding the minimum cost assignment that satisfies all demand. We use CO₂ emissions rather than traveled distance as our cost metric, since emissions better capture true costs by reflecting fuel consumption directly. We denote with $C_{l,o}^v$ the average CO₂ emissions for a ship in vessel class v traveling between locations l and o belonging to the set of locations $\mathcal{L}$ in the network.¹¹ We let T be the number of 5-day

⁹This is equivalent to imposing that load and discharge times are less than five days. [Figures 6](#) and [7](#) illustrate that this holds for the vast majority of voyages.

¹⁰The first assumption is justified by data and industry practice: all vessels sail at approximately the same speed (research in maritime engineering establishes the most economical speed for tankers, e.g., [Psaraftis and Kontovas \(2013\)](#)). The second assumption reflects the physical and economic constraints that segment the tanker market by vessel class. In particular, port infrastructure imposes draft and berth limits that restrict access for larger vessels. Also, cargo parcel sizes are matched to vessel capacity, making cross-class substitution economically impractical. Finally, waterway restrictions further reinforce this segmentation.

¹¹Emissions estimates are based on fuel consumption data that vessels are obligated to log to comply with the International Maritime Organization (IMO) regulations. The mapping from fuel consumption to emissions is an

Parameter	Explanation
$D_{o,d,t}$	Number of loads from area o to area d that depart in period t as observed in the dataset.
$\chi_{l,\tau,o,t}$	Indicator taking value 1 when a vessel departing location l at time τ reaches location o at t (clearly $\chi_{l,\tau,o,t} = 0$ if $\tau > t$). It is obtained by estimating the average ballast time between the two areas.
$I_{l,t}$	Number of vessels that become available in location l at time t , either for the first time (i.e., the first they are observed in the data) or after a prolonged stop (i.e., longer than the 90th percentile of the distribution of port stops).
$O_{l,t}$	Number of ships that cease to be available when in location l at time t , either because they exit the dataset or because of a prolonged pause.
$A_{d,t}$	Number of vessels that conclude a voyage in area d at time t .
$C_{l,o}^v$	Average CO ₂ emissions in tons of a ballast travel between areas l and o undertaken by a ship belonging to vessel class v .

Table 2: List of parameters estimated from the data and used in the optimization problems of Sections 3.1 and 3.2

periods from January 1st, 2018 to January 1st, 2020.

We introduce two types of decision variables: $X_{l,o,d,t}$, which counts the number of vessels that start their voyage at time $t \in \{0, \dots, T\}$ and location $l \in \mathcal{L}$ to serve a load picked up at origin $o \in \mathcal{L}$ and dropped at destination $d \in \mathcal{L}$; and $Y_{l,t}$, which denotes the number of vessels in location $l \in \mathcal{L}$ that remain unassigned at time t .

The formulation uses the following parameters obtained from the data: $D_{o,d,t}$ represents the number of loads departing from location o towards location d at time t ; $A_{l,t}$ represents the number of vessels that discharge at location l at time t ; ¹² $I_{l,t}$ represents the number of vessels that first enter the data in location l at time t ; and $O_{l,t}$ represents those vessels that leave the dataset after discharging in location l at time t . The indicator $\chi_{l,\tau,o,t} \in \{0, 1\}$ encodes whether a vessel leaving location l at time τ arrives in location o by time t , thereby capturing the average duration of a ballasting leg between the two locations. Further details about these parameters are in Table 2.

Using these parameters and variables, we state the integer program as:

$$\min \sum_{l,o,d,t} C_{l,o}^v X_{l,o,d,t} \quad (1a)$$

$$\text{s.t.} \quad \sum_{l,\tau \leq t} X_{l,o,d,\tau} \chi_{l,\tau,o,t} = D_{o,d,t}, \quad \forall (o,d,t) \in \mathcal{L} \times \mathcal{L} \times \{1, \dots, T\}, \quad (1b)$$

industry standard and takes into account the type of fuel used and the duration of use.

¹²Since we enforce that all loads must be transported, and the program is always feasible by construction, arrivals can be taken as exogenous to the decision variables.

$$\sum_{o,d} X_{l,o,d,1} + Y_{l,1} = I_{l,1}, \quad \forall l \in \mathcal{L}, \quad (1c)$$

$$\sum_{o,d} X_{l,o,d,t} + Y_{l,t} + O_{l,t} = A_{l,t} + I_{l,t} + Y_{l,t-1}, \quad \forall l \in \mathcal{L} \forall t = 1, \dots, T, \quad (1d)$$

$$X_{l,o,d,t}, Y_{l,t} \in \mathbb{N}. \quad (1e)$$

Equation (1a) is the total ballasting cost associated with the assignments in X . The constraints in Equation (1b) ensure that all loads observed in the data are fulfilled under the optimal assignment, and that the solution respects travel time feasibility. Constraints Equations (1c) and (1d) ensure that each vessel is used for at most one voyage at a time, and that the flow of ships at each location is conserved. On the right-hand side of each flow constraint, total supply at location l prior to the decision at time t consists of ships that remained in l from the previous period, newly entering vessels, and tankers arriving from voyages ending at l . On the left-hand side, supply is allocated to load assignments ($\sum X_{l,o,d,t}$), waiting ($Y_{l,t}$), or outflows ($O_{l,t}$). Without loss of generality, this formulation does not reposition empty vessels ahead of their assignment to loads.¹³

Let C_v^* be the optimal value of Equation (1a) for vessel class v and C_v^{obs} be the ballasting emissions measured for the same vessel class in the dataset. We define the share of ballasting due to trade imbalances as the ratio $Share_v^{Trade} = C_v^*/C_v^{obs}$. Table 3 reports the results, showing that the share of emissions due to trade imbalances ranges from 72% for smaller vessel classes to almost 90% for larger ones. This gradient is intuitive: larger vessels serve a limited set of routes, fewer ports can accommodate them, and the large loads destined to VLCCs follow more predictable patterns, leaving limited scope for network optimization. Conversely, Table 3 also reveals that factors beyond trade imbalances are relatively more important for smaller vessel classes, indicating significant potential for efficiency gains in those markets.

3.2 Fragmentation and Uncertainty

To isolate the effects of market fragmentation from uncertainty, we propose a comparative framework with three scenarios. At one extreme is the clairvoyant central planner who optimizes across all vessels with perfect foresight, representing the theoretical minimum ballasting cost C_v^* from the previous section. At the other extreme lie the observed data, which reflect the full impact of both fragmentation and uncertainty. Between these, we introduce an intermediate benchmark consisting of decentralized planners (one per commercial operator), each endowed with perfect information about their own fleet's future loads but acting independently. This intermediate case retains the inefficiencies of market fragmentation while removing the element of uncertainty. Comparing these

¹³Since there is no randomness in the problem, for every optimal solution that would have repositioned a tanker beforehand, there exists another that has the vessel ballasting to the origin of the same load just before the allocation.

Vessel class	Weight of trade imbalances (%)	No. vessels
MR1	77.14	357
MR2	74.49	1020
Panamax	73.76	308
Aframax	72.50	792
Suezmax	80.61	547
VLCC	88.34	756

Table 3: Share of ballasting attributed to intrinsic trade imbalances, obtained by comparing the current assignments against the assignments of a central planner matching tankers to loads in hindsight. The total number of vessels included in the optimization is 3,780. This figure is lower than the total number of vessels observed, because some entered the market after January 1, 2020.

three scenarios allows us to attribute inefficiencies quantitatively to either market structure or information limitations.

For each commercial operator $i \in \mathcal{P}_v$, where $\mathcal{P}_v$ denotes the set of operators in vessel class v , we solve the following problem.

$$\min \sum_{l,o,d,t} C_{l,o}^v X_{l,o,d,t}^i \quad (2a)$$

$$\text{s.t.} \quad \sum_{l,\tau \leq t} X_{l,o,d,\tau}^i \chi_{l,\tau,o,t} = D_{o,d,t}^i, \quad \forall (o,d,t) \in \mathcal{L} \times \mathcal{L} \times \{1, \dots, T\} \quad (2b)$$

$$\sum_{o,d} X_{l,o,d,1}^i + Y_{l,1}^i = I_{l,1}, \quad \forall l \in \mathcal{L}, \quad (2c)$$

$$\sum_{o,d} X_{l,o,d,t}^i + Y_{l,t}^i + O_{l,t}^i = A_{l,t}^i + I_{l,t}^i + Y_{l,t-1}^i, \quad \forall l \in \mathcal{L}, \forall t \quad (2d)$$

$$X_{l,o,d,t}^i, Y_{l,t}^i \in \mathbb{N} \quad (2e)$$

The parameters have the same interpretation as in [Section 3.1](#), with the difference that demand $D_{o,d,t}^i$, inflows $I_{l,t}^i$, and outflows $O_{l,t}^i$ are now computed at the commercial operator level. Let C_v^i denote the optimal value of this program, and define the system-wide ballasting cost of the decentralized system as $C_v^P = \sum_{i \in \mathcal{P}_v} C_v^i$. By construction, $C_v^* \leq C_v^P \leq C_v^{obs}$, so that

$$Share_v^{Trade} = \frac{C_v^*}{C_v^{obs}} \leq \frac{C_v^P}{C_v^{obs}} \leq 1.$$

The following two ratios then estimate the share of empty miles due to fragmentation and uncertainty, respectively:

$$Share_v^{Frag} = \frac{C_v^P - C_v^*}{C_v^{obs}} \quad (3)$$

Vessel class	Weight of uncertainty (%)	Weight of fragmentation (%)
MR1	10.68	12.18
MR2	13.41	12.1
Panamax	10.75	15.49
Aframax	11.02	16.48
Suezmax	9.07	10.32
VLCC	4.71	6.95

Table 4: Share of ballasting attributed to uncertainty and fragmentation

$$Share_v^{Uncertainty} = 1 - \frac{C_v^P}{C_v^{obs}}. \quad (4)$$

The numerator of Equation (3) compares the optimal cost under a centralized planner with no uncertainty against the cost of multiple independent fleets, also with no uncertainty; the only difference is the level at which assignments are made, so this difference captures precisely the increase in cost due to fragmentation. In Equation (4), the only difference between C_v^P and C_v^{obs} is that the former is calculated when vessel managers have perfect knowledge of the future; the gap therefore measures the extent to which uncertainty affects ballasting. We report the results for each vessel class in Table 4.

With the exception of VLCCs, the share of ballasting attributed to uncertainty is broadly similar across vessel classes, consistent with the intuition from Section 3.1 that VLCCs follow more predictable schedules due to the constraints imposed by their size. The more striking finding is that fragmentation accounts for a larger share of ballasting than uncertainty in every vessel class. This indicates that market structure is an important driver of carbon emissions, and that consolidating the control structure can yield substantial benefits both to the environment and to shipowners.

The case of the Aframax vessel class is particularly instructive. These medium-sized vessels are suitable for both long trans-oceanic voyages and shorter trips, and are capable of entering most ports and waterways; for these reasons, it is the fastest-growing vessel class among oil tankers and its popularity is expected to endure.¹⁴ Market fragmentation accounts for 16.5% of the carbon emissions from empty miles traveled by Aframax vessels, and since ballasting constitutes about 45% of total emissions for this class, fragmentation explains approximately 7.5% of all Aframax carbon emissions. For comparison, Pereda et al. (2025) estimate that imposing a \$100/tCO₂ carbon tax on international shipping would decrease emissions by only 3.6%. Addressing market fragmentation may thus yield larger near-term emission reductions than the leading carbon pricing proposals for oil shipping. That said, consolidating all tankers under a single entity is neither feasible nor desirable.

¹⁴See, e.g., <https://www.loydslist.com/LL1150330/Suezmax-and-aframax-segments-lead-tanker-newbuildings-as-orderbook-surges>.

[Section 4](#) examines shipping pools as a practical mechanism for achieving partial consolidation and reducing ballasting while preserving the competitive nature of the market.

3.3 Model Discussion

Our results imply that the carbon costs of market fragmentation are substantially higher than previously understood for the oil transportation market, and that efforts aimed at improving the sector’s environmental sustainability must consider their effect on market structure. While our analysis draws on data from the oil tanker market, the methodology can be applied with little modification to other decentralized transportation markets, such as dry bulk shipping and full-truckload trucking, that are likely to exhibit similar inefficiencies.

It is important to clarify the nature and scope of our estimates. Our decomposition should not be interpreted as a market counterfactual comparing ballasting costs under different equilibrium market structures. A full counterfactual analysis would require estimating the equilibrium response of transportation demand to consolidation and modeling the effect of consolidation on load fixing; it would also require replacing our cost-minimizing formulation with a profit-maximizing one. Estimating such a structural model is beyond the scope of this paper given the absence of data on contracted rates.¹⁵

We also note that holding demand fixed serves an important methodological purpose. A structural counterfactual that endogenizes demand would conflate two distinct channels: coordination gains from improved vessel assignment (our object of interest) and demand-side adjustments to changes in pricing and market power. By fixing demand, our approach cleanly identifies the first channel. This is analogous to standard decomposition exercises in which one factor is varied while others are held constant to isolate its contribution. Relaxing the fixed-demand assumption would not sharpen the estimate of coordination gains, but would blur it. Furthermore, keeping the demand-side of the market fixed is well justified in this setting: demand for oil transportation is highly inelastic with respect to the price of transportation (elasticity ≈ -0.026 , as implied by [Anderl and Caporale \(2024\)](#)), and any degree of consolidation likely to be implemented in practice, e.g., as we discuss in [Section 4.1](#), is unlikely to meaningfully affect oil freight rates.¹⁶

¹⁵Any structural model of this market must also account for the fact that commercial operators dynamically optimize a set of discrete resources, which yields a dynamic programming problem for which only approximate solutions exist even in the monopolistic setting (e.g., [Godfrey and Powell \(2002a\)](#)), while the oligopoly version is intractable. For these reasons, the literature on decentralized transportation markets typically assumes a collection of independent, non-atomic agents (e.g., [Brancaccio et al. \(2020\)](#)); however, this approach would fail to capture the benefits from consolidation.

¹⁶Specifically, [Anderl and Caporale \(2024\)](#) estimates that if the price of shipping refined oil increased by 29%, oil consumption in net importing countries would decrease by 0.75%. In other words, the elasticity of demand of oil to shipping costs in net oil-importing countries is about -0.026. As additional evidence, note that in 2019, the cost of shipping a barrel of oil from the Middle East to China was approximately 1% of the Brent crude index.

Finally, minimizing CO₂ emissions from ballasting, taking fulfilled demand as given, is closely aligned with minimizing costs: fuel constitutes by far the largest component of variable voyage costs, and in a market as fragmented as this one, individual operators have limited pricing power, making cost minimization a reasonable approximation to profit maximization.¹⁷ We therefore expect that neither the relative shares attributed to each factor nor the qualitative insights that follow in [Section 4](#) would differ substantially in a structural model with profit-maximizing agents.

Robustness check. Our treatment of uncertainty is “residual” in the sense that it is computed as the complement to trade imbalances and fragmentation. A potential concern is that this approach may overstate the role of uncertainty by combining two distinct types of randomness. The first is intrinsic uncertainty about the timing and location of future loads, which would persist even if all tankers were controlled by a single entity. The second concerns vessel-load assignments: even if transportation requests arrived deterministically, the outcome of the fixing process (which vessel ultimately transports a given load) would remain uncertain under the current market structure. This latter form of uncertainty is better understood as a by-product of fragmentation, because it would vanish under a centralized planner, and its effect on empty miles should therefore be attributed to fragmentation rather than to inherent demand uncertainty.

This distinction does not affect our estimate of the share due to trade imbalances in [Section 3.1](#), because the central planner can allocate any load in the data to any feasible vessel. However, in [Section 3.2](#) we assume that each operator’s clairvoyant planner knows exactly which loads it will be assigned, whereas a more realistic assumption would grant it information only about which loads will be *offered* to its fleet. To refine the treatment of the uncertainty component, [Appendix A](#) isolates the value of foresight in an online assignment environment by having a central planner solve a stochastic dynamic program using the state representation of [Godfrey and Powell \(2002a,b\)](#) and an approximate value function obtained via reinforcement learning. Given that this exercise holds market structure fixed and removes only the information advantage of perfect foresight, its uncertainty estimate serves as a lower bound that is nested within the broader three-way decomposition reported above. For tractability, we restricted this analysis to the Aframax vessel class. The stochastic approach estimates the share of ballasting due to uncertainty at 7% of observed costs, compared to 11% from [Table 4](#). The implied share of ballasting due to fragmentation for the Aframax segment is correspondingly bounded between 16% ([Table 4](#)) and approximately 20% ([Appendix A](#)). Although we cannot further narrow this range without observing the set of commercial operators to which each load was offered, the fact that two substantially different methodological approaches yield estimates in close agreement provides confidence in the robustness of our findings.

¹⁷See, e.g., [Brancaccio et al. \(2023a\)](#) who argue that shipowners have limited ability to pass onto their customers variations in the cost of fuel.

A final related concern is that single-ship operators cannot benefit from perfect foresight in our framework, potentially inflating our estimate of the fragmentation tax. We note, however, that single-ship operators account for fewer than 2% of voyages and 1-3% of ballasting emissions, so that excluding them does not materially affect our estimates. Moreover, the dynamic programming approach of [Appendix A](#), which estimates uncertainty keeping market structure fixed, yields a fragmentation share of 20% for Aframax vessels, which is consistent with our main estimates.

3.4 Market-Level Empirical Validation

The optimization-based approach of the previous sections provides a clean estimation of the drivers of ballasting in oil shipping. That market fragmentation accounts for such a large share of observed empty miles suggests that consolidating vessel control would reduce ballasting costs and emissions. In this section, we test this prediction by exploiting variation in market concentration across geographic areas and time. Our analysis confirms that consolidation reduces the inefficiencies we identified in [Section 3.2](#) and further validates our optimization-based approach.

Our empirical approach is to estimate the extent to which markets with higher concentration of tanker control exhibit higher efficiency. We measure ballasting efficiency along three dimensions: average ballast distance, average ballast times, and average CO₂ emissions. Starting from our voyage-level data, we construct a panel of “markets” (one market for each combination of load area and vessel class) observed across 16 quarters (28 areas, 6 vessel classes, 16 quarters). For each market we compute the average ballasting efficiency measures, as well as the Herfindahl-Hirschman Index (HHI) of operator concentration. The HHI is defined as the sum of squared market shares of each operator, where the market share of operator i in quarter t in a given load area-vessel class (l, v) is the share of loads originating from that load area during quarter t that are operated by i . We then estimate the following specification:

$$\log(\bar{Y}_{l,v,t}) = \beta \times \log(\text{HHI}_{l,v,t}) + \alpha_{l,v} + \xi_t + \varepsilon_{l,v,t}, \quad (5)$$

where $\bar{Y}_{l,v,t}$ is the market-average outcome in area l , vessel class v , and quarter t ; $\alpha_{l,v}$ is an area-by-vessel-class fixed effect; and ξ_t is a time (quarter) fixed effect. The OLS estimates of this regression are presented in [Table 5](#).

The concentration level of a market is potentially endogenous. One may expect reverse causality to be relevant in this setting: if ballasting is costly for certain load areas, operators may decide to avoid them, thus inflating the level of concentration. To address this concern, we instrument for $\text{HHI}_{l,v,t}$ using its one-quarter lag, $\text{HHI}_{l,v,t-1}$. Because we already include area-vessel class fixed effects, we only need to address endogeneity arising from shocks that make ballasting to a load area

temporarily less efficient. Given that ballasting trips are long, quarter-to-quarter HHI changes are driven by slow-moving entry and exit decisions that precede contemporaneous ballast outcomes. The lagged HHI is thus a *strong* and *valid* instrument; the first-stage F-statistic in Table 5 confirms this. As a robustness check, we also employ a Bartik-style shift-share instrument that interacts predetermined operator market shares from four quarters prior with exogenous global trends (we report the latter results in Appendix B).

	Ballast Distance (nm)		Ballast Emissions (CO ₂)		Ballast Time (days)	
	OLS	Lagged HHI IV	OLS	Lagged HHI IV	OLS	Lagged HHI IV
<i>Independent Variable</i>						
Log (HHI)	-0.2581*** (0.0543)	-0.3229*** (0.0667)	-0.2428*** (0.0461)	-0.3039*** (0.0591)	-0.2026*** (0.0458)	-0.2434*** (0.0527)
<i>Fixed-effects</i>						
Load Area	Yes	Yes	Yes	Yes	Yes	Yes
Quarter	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	2,297	2,143	2,297	2,143	2,297	2,143
R ²	0.90674	0.99175	0.94668	0.99397	0.93063	0.99455
Wald (1st stage F-stat)	N/A	100.85	N/A	100.85	N/A	100.85
<i>Clustered standard-errors in parentheses</i>						
<i>Signif. Codes: ***, 0.01, **, 0.05, *, 0.1</i>						

Table 5: Market-level regressions of ballast efficiency measures on market concentration (HHI). The instrument for the Lagged HHI IV specification is $\log(\text{HHI}_{l,v,t-1})$. Similar results with Bartik-style IVs are presented in Appendix B.

Notes: Standard errors clustered at the Load Area $\times$ Vessel Class level. Regressions weighted by number of trips. From the total sample of $28 \times 6 \times 16 = 2,688$ market-quarter pairs, we restrict attention to those with at least 10 departing loads, yielding 2,297 market-quarter observations. For IV regressions, sample size is smaller due to $t = 1$ markets not having a well-defined IV.

Table 5 presents the results. All coefficients are negative and statistically significant, confirming our findings from the previous sections: when the transportation market is more concentrated, ballasting is more efficient. The IV estimates imply that a 1% increase in market HHI is associated with a 0.32% decrease in average ballast distance and a 0.30% decrease in ballast CO₂ emissions. To gauge the economic magnitude, consider moving from the 25th percentile of HHI in the panel (0.065, highly fragmented) to the 75th percentile (0.173, moderately concentrated): this implies a 26% reduction in ballast distance and a 25% reduction in ballast CO₂. A doubling of HHI is associated with a 19% reduction in both distance and emissions. These magnitudes are broadly consistent with our estimates from Sections 3.1 and 3.2, where full consolidation yields a 15% reduction in ballast emissions (Table 4). The model computes an upper bound under perfect foresight, whereas the regressions in this section capture realized gains across observed HHI variation, including real-world frictions. The consistency between the two approaches provides empirical validation of the

model’s core prediction.

Finally, we note that observed HHI variation in the panel is modest: the median is 0.10, reflecting the high fragmentation documented in [Section 2](#). Our results so far point to the possibility of substantially improving the efficiency of this transportation market by consolidating vessel control, which naturally raises two questions: what level of consolidation is needed to capture the bulk of the potential efficiency gains, and what mechanisms would drive this improvement in practice. We address these questions in the next section.

4 Shipping Pools and Efficiency Gains

In the previous section, we established that market fragmentation is responsible for a significant share of the costs incurred by the oil transportation market, accounting for 7-16% of all empty miles depending on vessel class. We derived these estimates by comparing observed outcomes against two benchmarks: a first-best benchmark in which a single clairvoyant planner controls the entire fleet, and a second benchmark in which there are as many independent planners as there are commercial operators in the data. We further validated our analysis empirically by exploiting variation in vessel control across different geographic markets, and our conclusion is that consolidation of this market would improve efficiency. However, while informative as an upper bound, full centralization is neither feasible nor desirable: it would eliminate competition and raise obvious regulatory concerns. This section asks whether a more pragmatic form of consolidation, one that preserves the market’s competitive structure, can capture a substantial share of the first-best gains, and what factors would drive this improvement.

Shipping pools are a natural candidate for consolidating vessel control. Under a pool agreement, independent shipowners place their vessels under unified commercial management and share the revenue generated by the combined fleet. Pools are an established feature of the oil tanker industry, and their adoption has accelerated in recent years: between 2023 and 2024 (outside our data time horizon) alone, several new pools were formed across vessel classes, including the Hafnia-Mercuria Panamax pool and the Scorpio-Mercuria “Mercury Pool” for Aframax vessels, while existing pools such as Tankers International expanded further. Because shipowners retain the flexibility to enter or exit a pool at any time, antitrust regulators have generally held a more permissive view of pooling arrangements compared to outright mergers or codified shipping alliances, making pools unlikely to attract the regulatory push-back that accompanies other forms of consolidation.

Our analysis of shipping pools proceeds in two parts. We first study the potential for *partial* consolidation to improve market-level efficiency, asking how the size of the pool affects total ballasting and whether high concentration is necessary to substantially reduce empty miles. We then

turn to the micro-level mechanisms through which larger pools achieve these gains, examining how fleet size affects vessel positioning, route selection, and load assignment.

4.1 Partial Consolidation

Our dataset contains 169 Aframax commercial operators, and in [Section 3.1](#) we evaluated the inefficiency of the current market by solving for the assignments of a single entity controlling all vessels. To measure the value of *partial* consolidation, we create “synthetic” shipping pools of varying size. This allows us to test different degrees of consolidation and assess the trade-off between coordination gains and market concentration.

To track the efficiency gains from consolidation we use as our metric the *share of first-best ballasting savings* achieved by a given consolidation level. Formally, let the first-best ballasting savings Δ_v^* for vessel class v be defined as $\Delta_v^* = (C_v^* - C_v^{obs}) / C_v^{obs}$, where C_v^* is the optimal cost from [Equations \(1a\) to \(1e\)](#) and C_v^{obs} is the observed ballasting emissions.

We randomly partition the tankers into shipping pools of identical size ϕ , where ϕ denotes the fraction of the total number of vessels that each pool manages. For each pool p , we observe the voyages performed by its vessels and solve an optimal assignment problem in the form of [Equations \(2a\) to \(2e\)](#), yielding an optimal cost $C_{p,v}^*(\phi)$. The market-wide ballasting cost is then $C_v^*(\phi) = \sum_p C_{p,v}^*(\phi)$ and the gain from ϕ -consolidation is denoted $\Delta_v^*(\phi)$.¹⁸ Our metric of interest is the ratio $\rho_v(\phi) = \Delta_v^*(\phi) / \Delta_v^*$, which reports the fraction of the first-best savings achieved when the market is consolidated into $\frac{1}{\phi}$ pools each controlling a share ϕ of vessels.

The left panel of [Figure 1](#) plots $\rho_v(\phi)$ for different vessel classes and pool sizes, showing how much of the fragmentation inefficiency can be eliminated at varying levels of consolidation. The right panel illustrates the decrease in average ballast time as the level of consolidation increases. These figures establish that even modest-sized pools, each comprising approximately 5% of the market, capture the vast majority of the first-best savings. This is particularly evident for smaller vessel classes such as Aframax or MR2: partitioning all 1,146 Aframax ships into pools of 60 units each (about 5% of the fleet) yields between 80% and 90% of the savings that a centralized planner would achieve. Using our estimates from [Table 4](#), this corresponds to an overall decrease in ballasting emissions of about 15%, or 4.6 million tonnes of CO₂ not released into the atmosphere annually (equivalent to removing approximately one million passenger cars from the road).

The shape of these curves indicates sharply decreasing marginal returns to consolidation, reminiscent of the analysis on flexibility designs initiated by [Jordan and Graves \(1995\)](#) despite the different context. Larger fleets provide greater flexibility for a pool manager to optimize ballasting destinations and load assignments, but most of this flexibility is exhausted well before a pool

¹⁸By definition, $\Delta_v^* = \Delta_v^*(1)$, since it is obtained with one fleet controlling 100% of the vessels.

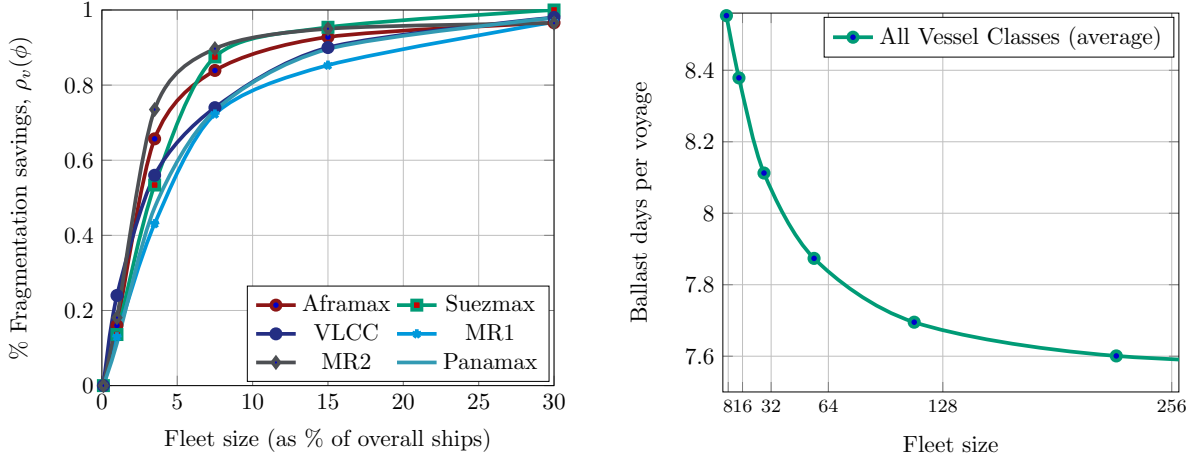


Figure 1: Effects of partial consolidation. Left: share of fragmentation inefficiencies recouped by pooling vessels into fleets of different sizes. Right: average length of the ballast leg (in days) as the size of synthetic fleets increases (log axis); we show a weighted average (by number of voyages) across vessel classes for readability.

reaches even a modest share of total market capacity. This is an important finding from a policy perspective: the level of consolidation sufficient to capture the bulk of coordination gains is far too low to confer meaningful market power.

4.2 Consolidation and Operator-Level Efficiency Gains

The results of the previous subsection establish *that* partial consolidation reduces ballasting, but they do not reveal *how*. The synthetic pool exercise re-optimizes vessel-load assignments holding everything else fixed, which isolates the coordination channel but leaves open the question of which operational mechanisms drive the gains in practice. We now turn to the observed behavior of real commercial operators to investigate these mechanisms, shifting from market-level aggregates to the micro-level decisions of individual pools.

Our measures of interest change accordingly. While the previous sections focused on total ballasting emissions and aggregate fragmentation, we now concentrate on the *size* of individual shipping pools (i.e., the number of vessels a commercial operator controls), how it varies over time, and how it influences operators' choices.

In the following sections we carry out several regression analyses, where the primary independent variable is the *weighted pool size*, defined at the commercial operator level. For any given period (e.g., a quarter) we calculate pool size as a weighted count of vessels under the operator's management during that period. This calculation accounts for the number of days each vessel

spends operating (whether ballasting, laden, or at port) for the commercial operator, relative to the total number of days in the period:

$$PoolSize_{i,t} = \sum_s \frac{T_{s,i,t}}{T_t},$$

where $T_{s,i,t}$ equals the number of days in period t that vessel s spent under commercial operator i and T_t is the total number of days in period t . This approach provides a more nuanced representation of an operator’s fleet capacity over time, accounting for fleet changes occurring mid-period. For instance, if an operator managed two ships in April 2018, with one operating for the entire month and another joining mid-month, the calculated pool size would reflect this partial utilization.

Addressing endogeneity. Using pool size as a covariate raises potential endogeneity concerns, since the decision to increase the number of vessels under control is partly influenced by the profitability of a shipping pool. There are two main risks: one stems from time-varying unobservable characteristics of commercial operators that affect both efficiency outcomes and pool size; the other is reverse causality. As far as the first is concerned, we view pool size as a proxy for an operator’s overall “market weight,” encompassing precisely these correlated unobservable factors. However, reverse causality is more worrisome: an operator might first improve efficiency (e.g., through investments in analytical capabilities), subsequently attracting more vessels to its pool. We address this risk using instrumental variables for the pool size of operator i during period t . Motivated by the overall growth in the number of oil shipping vessels over our time horizon, we choose two Hausman-style IVs: in both cases we use as the IV for the pool size of operator i at time t the average pool size within a “comparable” group of operators other than i .¹⁹ We bin operators at vessel class level by pool size quartiles and geographical focus, and then define the two IVs as: (i) the average number of ships in the same vessel class that operators in the same quartile as operator i and with geographical focus different from i control; (ii) the average number of ships in all other vessel classes that operators in the same quartile as operator i and with geographical focus different from i control. We call the first instrumental variable *SameVC*, and the second *CrossVC*. Detailed definitions can be found in [Appendix B](#).

Shipping pool utilization. The most commonly used measure of efficiency for a transportation company is *utilization*, computed as the ratio between how much a resource was used for productive purposes and total usage. Because managers of shipping pools focus on the overall efficiency of the pool rather than individual vessels, we define utilization *at the pool level*. Formally, let time-

¹⁹Building oil tankers takes up to three years, meaning that new vessels entering service in, e.g., 2020 were ordered well before any potentially correlated shock to pool efficiency.

utilization $Utilization_{i,t}^T$ for operator i during period t be defined as the ratio between the total time spent laden and the total duration of voyages beginning in period t , i.e.,

$$Utilization_{i,t}^T = \frac{\sum_n T_{n,i,t}^L}{\sum_n T_{n,i,t}^V},$$

where $T_{n,i,t}^L$ denotes the duration of the laden leg of the n -th voyage operated by commercial operator i whose start date is in period t , and $T_{n,i,t}^V$ denotes the total travel time of the same voyage. Note that the definition imputes the utilization level based on the starting date of the voyage, even if, because of long travel times, the pool size may have changed by the time the tanker secures another contract. Since we do not observe the fixing date of the loads, we assume that the decision about ballasting destinations is made at the starting date of the voyage.

Utilization regression. With this definition of utilization, we seek to understand whether an operator becomes more efficient when its pool size grows. Specifically, we construct a monthly panel of operators' efficiency metrics and estimate the following regression:

$$\log(Utilization_{i,t,v}^T) = \beta \times \log(PoolSize_{i,t,v}) + \xi_t + \psi_{i,v} + \varepsilon_{i,t}, \quad (6)$$

where ξ_t is a month-level fixed effect and $\psi_{i,v}$ is an operator- and vessel class-specific fixed effect. Note that in this regression $PoolSize_{i,t,v}$ is the (time-weighted) number of vessels that operator i controls in vessel class v and $Utilization_{i,t,v}^T$ is similarly defined. The coefficient of interest is β , which represents the elasticity of utilization to changes in the pool size. The estimates are reported in [Table 6](#), both for OLS and IV regressions. The positive utilization effect confirms that the market-level gains documented in [Section 3.4](#) translate into operator-level benefits: larger operators spend a greater fraction of their operations carrying revenue-generating cargo rather than idling in search of their next load. To illustrate the magnitude of this effect, consider an operator scaling from the median fleet size of $N_{\text{median}} = 3.3$ vessels to a pool of $N_{\text{pool}} = 60$ vessels, a size that [Section 4.1](#) demonstrates is sufficient to capture roughly 85% of the total available coordination gains. The log-log specification implies a change in log utilization of

$$\Delta \log(U) = \hat{\beta} \times \log\left(\frac{N_{\text{pool}}}{N_{\text{median}}}\right) \approx 0.018 \times \log\left(\frac{60}{3.3}\right) \approx 0.052,$$

which corresponds to a relative increase in utilization of approximately $e^{0.052} - 1 \approx 5.3\%$. Starting from a baseline median utilization of 56.8%, this implies a new utilization level of approximately 59.8%, and a corresponding decrease in the ballast share of voyage time from 43.2% to 40.2%. The relative reduction in the ballast share is therefore approximately $3/43.2 \approx 7\%$. Because vessels

within each class travel at approximately the same speed, reductions in ballast time translate proportionally into reductions in ballast distance and emissions. Since a 60-vessel pool captures about 85% of the theoretical maximum coordination gains, the implied total potential reduction in ballasting is approximately $7\%/0.85 \approx 8.2\%$, which falls within the range of the fragmentation share estimated in Table 4 (7–16% depending on vessel class). The fact that this regression-based estimate lands at the lower end of the range is to be expected: the estimates from Sections 3.2 and 3.4 account for improvements that occur both *within* each pool and *across* pools at the equilibrium level; on the other hand, Equation (6) focuses on increased efficiency at the operator level.

Dependent Variable: Model:	OLS	Time-utilization	
		IV Same VC	IV Cross VC
<i>Variables</i>			
$\log(PoolSize)$	0.0184*** (0.0040)	0.0173** (0.0071)	0.0176** (0.0069)
Granularity	Month	Month	Month
<i>Fixed-effects</i>			
Operator $\times$ Vessel Class	Yes	Yes	Yes
Month	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	24,455	24,455	24,455
R ²	0.28659	0.28536	0.28537
Wald (1st stage F-stat)	N/A	310.76	297.26
<i>Clustered (Commercial Operator) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Table 6: Operator-level regressions of average pool utilization over the pool size. Additional specifications with a distance-based definition of utilization presented in Appendix B.

While these results confirm that larger pools are more efficient, utilization is a summary measure that does not reveal *which* operational mechanisms drive the improvement. The next section examines two specific channels through which pool size affects fleet management decisions.

4.3 Pool Size, Ballast Delays, and Port Reach

We have argued that consolidating vessel control benefits both the market as a whole and individual commercial operators, which become more efficient. In this section, we study how these micro-level improvements arise, focusing on how pool size affects the way operators manage their fleets. Specifically, we shift to voyage-level data and concentrate on two complementary dimensions: the average time it takes for a vessel to find a new load, and the number of ports that vessels visit

within a quarter. The former captures whether pool size affects how operators arrange loads, while the latter captures how structural changes in the network of locations served drive efficiency.

Ballast delays. As we know from [Section 3.1](#), a certain amount of ballasting is unavoidable. Moreover, under a central coordinator a given oil tanker might be required to travel longer ballast legs if this benefits all the vessels in the pool. Taken together, these observations imply that we should not expect a generalized decrease in total ballast time for every voyage operated by pools whose fleet grows. Instead, to uncover how pool size benefits efficiency, we focus on how quickly oil tankers get assigned a new load *once they have been given a ballast route*. This approach allows us to look deeper into the operations of shipping pools, beyond the more coordinated load-vessel assignment that we expect from [Section 3.2](#). To do so, we compare the travel time of the ballast leg of each voyage against a “non-delayed” benchmark for the same route, defined as the 20th percentile of all observed travel times, analogous to the “free-flow” travel time concept in the transportation literature. This allows us to abstract from delays due to factors such as port congestion. We define the following variable for each voyage:

$$BallastDelay_{n,i,t} = TravelTimeBallast_{n,i,t} - NonDelayedTimeBallast_n$$

where $BallastDelay_{n,i,t}$ captures the difference between actual and non-delayed travel time for the ballast leg of voyage n served by operator i at time t .

We then regress the extra ballasting time of each voyage on the pool size of the operator that served the load. As in [Section 4.2](#), we instrument the pool size using two Hausman IVs.

$$BallastDelay_{n,i,t} = \beta \times \log(PoolSize_{i,t}) + \phi_{o,d} + \xi_t + \psi_{i,v} + \varepsilon_{n,i,t}. \quad (7)$$

Here, $\phi_{o,d}$ denotes an origin-destination (route-specific) fixed effect, ξ_t denotes a time fixed effect, and $\psi_{i,v}$ denotes the commercial operator and vessel class fixed effect. The first three columns of [Table 7](#) report the estimates for the coefficient of interest. The OLS and 2SLS coefficients are negative and significantly different from zero, indicating that vessels belonging to larger pools take less time to secure a new load once they are on a ballast route. In particular, the IV estimates show that doubling fleet size is associated with a half-day reduction in ballast time per voyage. To put this in context, a fleet of ten tankers, each completing one voyage per month, would save five days of ballasting every month compared to two separate fleets of five vessels each; at average Aframax emissions of 60 tons of CO₂ per day, this amounts to 3,600 tons of CO₂ per year, equivalent to roughly 800 passenger cars. Since we include a route-specific fixed effect, the reduction in ballast time does not reflect differences in route composition. Instead, it is consistent with larger pools

having greater scheduling flexibility and a larger set of vessels across positions, enabling faster matching between available capacity and cargo opportunities.

Dependent Variable:	BallastDelay			Number of ports visited		
	OLS	IV SameVC	IV CrossVC	OLS	IV SameVC	IV CrossVC
<i>Independent Variables</i>						
$\log(PoolSize)$	-0.2152** (0.1050)	-0.5424** (0.2440)	-0.4699** (0.2251)	0.2547*** (0.0114)	0.2317*** (0.0269)	0.0975*** (0.0332)
<i>Fixed-effects</i>						
Commercial Operator-VesselClass	Yes	Yes	Yes	Yes	Yes	Yes
Period	Yes	Yes	Yes	Yes	Yes	Yes
Starting-Load port pair	Yes	Yes	Yes	—	—	—
IMO	—	—	—	Yes	Yes	Yes
Geographic granularity	Area	Area	Area	Port	Port	Port
Time Granularity	Month	Month	Month	Quarter	Quarter	Quarter
<i>Fit statistics</i>						
Observations	175,509	175,509	175,509	78,313	71,215	72,488
R ²	0.11680	0.11679	0.11679	0.46275	0.45601	0.45095
Wald (1st stage F-stat)		127.09	139.86		208.55	145.43

Clustered (Commercial Operator) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 7: Voyage-level regressions of the extra ballasting time and number of ports visited by a vessel on pool size. Similar results hold at different geographical and time granularities and are presented in [Appendix B](#).

Notes: Because imposing fixed effects for each port pair exponentially increases the number of variables to estimate, with resulting loss of statistical power, we estimate origin-destination fixed effects at area level. However, we compute travel times at port level (i.e., if two legs are between different ports that belong to the same origin/destination area, their travel times will be different).

Port reach. The magnitude of the ballast delay reduction suggests that faster load-securing alone cannot fully account for the increased utilization we observe. We now examine another observable dimension of operator behavior that underpins overall pool efficiency: the network of locations served by each fleet. Large commercial operators are naturally expected to serve more locations simultaneously, so we carry out a vessel-level analysis to ensure meaningful comparisons. Specifically, we focus on the number of distinct ports a given vessel visits over a quarter. Formally, we estimate the following specification:

$$\log(No.Ports_{s,i,t}) = \beta \times \log(PoolSize_{i,t}) + \xi_t + \psi_{i,s} + \varepsilon_{s,i,t}. \quad (8)$$

Here, $No.Ports_{s,i,t}$ counts the distinct ports visited by tanker s in quarter t while under operator i 's management, ξ_t is the quarter fixed effect, and $\psi_{i,s}$ is a vessel- and commercial operator-specific

fixed effect. Thus, this regression compares *the same vessel* as it moves from one operator to another and the pool size it coordinates with changes; in other words, Equation (8) examines whether vessels in larger fleets are deployed more extensively across the network. We instrument $PoolSize_{i,t}$ using the same two Hausman IVs as before, with the proviso that the IVs are now calculated at quarterly level.

The last three columns of Table 7 show the estimates for the effect of pool size on the number of ports visited by a vessel: all coefficients are positive, indicating that the same ship visits a wider range of locations within a quarter when managed by a larger operator. With an estimated elasticity of 0.23, doubling an operator’s fleet size allows its individual vessels to expand their operational network by approximately 17%. In other words, smaller pools concentrate on a few routes, traveling the same paths repeatedly, while larger pools deploy the same vessels across more ports and more diverse loads.

These findings point to two reinforcing mechanisms behind the coordination gains documented in Sections 4.1 and 4.2. First, larger pools achieve shorter ballast delays, reflecting greater scheduling flexibility: with more vessels distributed across locations, a pool is more likely to have capacity positioned near a cargo opportunity when it arises, reducing matching time. Second, the broader geographic reach of larger pools provides diversification against local demand fluctuations and reduces dependence on individual trade lanes. Importantly, both mechanisms exhibit strong returns at modest scale, consistent with the concavity of the gains curve in Section 4.1: a pool controlling roughly 5% of fleet capacity already operates a sufficiently diversified network to capture most of the available coordination gains, while remaining too small to exercise meaningful market power.

5 Concluding Remarks

This paper provides a quantitative decomposition of how market structure, as distinct from geography and uncertainty, drives inefficiency in a large-scale transportation market. Applied to the global oil tanker industry, our framework reveals that fragmented ownership accounts for 7-16% of total empty miles depending on vessel class, a share that is roughly double the contribution of demand uncertainty. We call this the fragmentation tax: a cost, both financial and environmental, that is structural in nature and operationally addressable without technological change. Two complementary empirical approaches, i.e., optimization and instrumental variable regressions exploiting regional variation in market concentration, yield consistent estimates, strengthening confidence in this finding.

The fragmentation tax points to operational coordination as a practical, near-term complement to the technological and regulatory strategies that dominate the decarbonization agenda. Ship-

ping pools offer a mechanism for capturing these gains: consolidating vessels into pools controlling roughly 5% of a given vessel class captures 80-90% of the efficiency gains achievable under full centralization, reducing ballasting-related emissions by up to 15%. These gains arise at consolidation levels far too low to confer meaningful market power, and manifest through two reinforcing channels: greater scheduling flexibility that reduces ballast delays, and broader port networks that support route diversification. Unlike carbon pricing, which may yield only modest near-term reductions, or alternative fuels requiring trillions in investment and decades of fleet turnover, pool formation requires only modest changes to how existing assets are commercially managed.

From a policy perspective, several concrete measures could accelerate pool adoption. First, regulatory clarity would reduce a key barrier to entry: maritime antitrust authorities could issue explicit safe-harbor guidelines establishing that pooling arrangements below a defined market share threshold (e.g., 5-10% of vessel class capacity) are presumptively compliant with competition law, removing the legal uncertainty that currently discourages smaller owners from joining pools. Second, targeted tax incentives could align private and social returns to coordination. Because the environmental benefits of reduced ballasting are externalities that individual pool members do not fully capture, instruments such as reduced port fees or tonnage tax credits for pooled vessels would help close this gap. The EU Emissions Trading System, which already covers maritime shipping, provides a natural mechanism: pooled fleets that demonstrably reduce their emissions intensity could receive preferential allowance allocations or rebates, rewarding coordination directly within the existing regulatory architecture. Finally, industry bodies and classification societies could establish standardized pooling contracts and transparent performance benchmarks, reducing the transaction costs of forming new pools and the information asymmetries that make shipowners reluctant to place vessels under external management.

Several limitations of our analysis point to directions for future research. Our decomposition holds demand fixed, which cleanly isolates the coordination channel but does not capture potential demand-side responses to consolidation. In the oil tanker market, where demand for transportation is highly inelastic with respect to freight rates, endogenizing demand is unlikely to materially alter our estimates. However, a full structural equilibrium model that jointly determines pricing, demand, and fleet allocation would be the appropriate framework for markets with more elastic demand, e.g., full-truckload trucking. Developing such a model would require data on contracted rates, which are challenging to collect in most transportation markets, but would provide a more complete picture of the welfare implications of consolidation. Extending the analysis to study how operational coordination interacts with carbon pricing is another natural direction: if pools reduce baseline emissions, the marginal abatement cost curve for the industry shifts, potentially altering the optimal carbon price. Finally, while our empirical setting is maritime shipping, the methodology

applies to any market in which independent operators make spatial repositioning decisions, e.g., dry bulk shipping or trucking, where the magnitude of the fragmentation tax is an empirical question.

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Appendices

A Uncertainty: Approximate Dynamic Programming

Uncertainty is one of the main characteristics of the maritime world. In the case of oil transportation markets, shipowners and managers face uncertainty from three different sources: (i) from where new loads will be offered, (ii) when, and (iii) the level of competition for each of these (i.e., if they will outbid their opponents and thus win the load). The first two elements are unaffected by fragmentation in the market, but the third is: in a very fragmented market there are many competitors for each load, which decreases the likelihood that a given tanker will win it. With this appendix we aim at disentangling the share of ballasting due to “pure” uncertainty from the share due to uncertainty that can be mitigated by consolidating the market.

Our approach is similar to what we followed in [Section 3](#): we consider a central planner whose objective is to minimize the ballasting cost associated with transporting the loads. The main difference is that now we assume that the central planner does not have perfect foresight; instead, it has a probabilistic assessment about the distribution of future loads, and makes decisions based on this. Formally, the central planner solves a stochastic dynamic program. We then compare the optimal costs obtained in the DP with C_v^* from [Section 3.1](#); since the two problems only differ in terms of uncertainty about the future, the difference in optimal cost can be taken as a measure of the share of ballasting due to it. Moreover, since we are comparing ballasting costs incurred by two central planners, in both cases there is no uncertainty about competition. It follows that the measure we derive captures only pure uncertainty as discussed above.

Following discussions with our industry partner, we focus on the Aframax market, which has a number of attractive features. Aframax vessels are of intermediate size, so they can dock at most ports in the world and can use both the Panama Canal and Suez Canal. Moreover, an increasing number of them are suitable for transporting both crude oil and refined products, which makes it a segment expected to grow in popularity as MR1 and MR2 decrease their market share.

We approximate the optimal value function of the stochastic DP, denoted by $V^*(s)$, where s is a state summarizing the present and future availability of tankers. We then find the ballasting decisions that the central planner would have made if faced with the demand realizations observed in the data. Because the central planner optimizes *online*, we cannot ensure that all loads observed are transported. In our simulation we find that the central planner can satisfy 75% of the observed loads. We take a conservative stance and compare the minimum cost obtained with the approximate value function against a benchmark calculated as follows: for every route (o, d) and every time period t we sort the loads observed in reality in ascending order of ballasting emissions, and then

consider only the $n_{o,d,t}$ least costly, where $n_{o,d,t}$ is the number of loads on (o, d) in period t that were served in the simulation; we then sum the ballasting emissions associated with these voyages. Compared against this benchmark, the central planner achieves ballasting costs that are 20% lower than observed. This suggests that for the Aframax class, uncertainty accounts for about 7.5% of the overall ballasting cost, with the remaining 20% attributable to operational inefficiencies and uncertainty regarding competition. The remainder of the section presents the formal arguments needed to compute an estimate of V^* .

A.1 Dynamic Programming Formulation

Consider a decision maker that minimizes ballasting costs over discrete time periods $t = 0, 1, \dots$, that represent 5-day periods in the data. Demand from origin $o \in \mathcal{L}$ to destination $d \in \mathcal{L}$ at time t is denoted by $D_{o,d,t}$ and is drawn i.i.d. from a Poisson distribution with mean $\lambda_{o,d}$. As in the case of [Section 3](#), $\mathcal{L}$ is the set of geographical areas from [Table 16](#); let $N_A = |\mathcal{L}| = 28$. We estimate $\lambda_{o,d}$ as the average number of loads observed on the (o, d) route in the data.²⁰ Travel times between locations are deterministic, denoted by $T_{o,d}$, and equal to the average travel time observed in the data; let $T_{\max}$ denote the maximum length of a voyage in this world, i.e.

$$T_{\max} = \max_{l,o,d} T_{l,o} + T_{o,d}.$$

Given a total number of vessels equal to N , the state of the system at every time t is represented by $s^t \in S$. The state space S is finite, equal to the set of all $28 \times (T_{\max} + 1)$ matrices with natural entries that sum up to N . Formally,

$$S = \left\{ s \in \mathbb{N}^{28 \times (T_{\max} + 1)} : \sum_{d,\tau} s_{d,\tau} = N \right\}.$$

Column τ of state s^t is the number of ships that will become available in $\tau - 1$ periods in the future as a result of voyages begun in all periods up to $t - 1$ (inclusive) and that have not yet reached their destination. So, the first column represents the number of vessels currently available in each location, the second column the number and destination of vessels that will terminate their voyage in the next period, and so on. Let $s_{\cdot,\tau}$ denote the τ -th column of state s . Therefore, s^t summarizes the future availability of tankers given the decisions made until time $t - 1$.

At every time period t , the central planner observes the demand realization and decides how many loads to serve and how. Specifically, it acts on two decision variables: $X_{l,o,d}$ denotes the

²⁰Here it is clear why we need to consider only data before the outbreak of COVID-19. The pandemic substantially altered oil trade flows.

number of vessels available in l used to transport loads from demand $D_{o,d,t}$; $B_{l,d}$ represents the number of vessels ordered to ballast from l to d , with the convention that $B_{l,l}$ equals the number of vessels ordered to wait in l . We assume that the central planner can serve a load on route (o, d) at time t only with currently available tankers that can reach location o from their position by the same period t . Note that in this formulation a vessel may be ordered to ballast with a cargo already secured ($X_{l,o,d}$ for $l \neq o$), or in expectation of new loads in the future ($B_{l,d}$ for $l \neq d$). Based on these decisions, the state deterministically transitions to s^{t+1} : all travels scheduled to terminate in $\tau - 1$ periods in s^t are in column $\tau - 1$ in s^{t+1} ; and travels that take time T appear in column $T - 1$. In particular, s^{t+1} can be written as a linear function of s^t .

In [Section 3.1](#) we imposed the constraints of [Equation \(1b\)](#), which require that all loads be transported. This is possible because the problem is deterministic; for the stochastic DP at hand, we impose instead that the central planner suffers a penalty $M > 0$ for each load that remains unassigned. Together with the ballasting cost paid for relocations and assignments, we obtain a flow-payoff function

$$r_t(X, B|s, D) = \sum_{l,o,d} C_{l,o}^v X_{l,o,d} + \sum_{l,d} C_{l,d} B_{l,d} + M \sum_{o,d} \left(D_{o,d,t} - \sum_l X_{l,o,d} \right). \quad (\text{A.1})$$

The central planner seeks to minimize $\sum_{t=0}^{\infty} \gamma^t r_t$, where γ is a discount factor.²¹ It is well known that the optimal value starting from state s of this dynamic program, denoted $V^*(s)$, satisfies the Bellman equation, i.e.,

$$V^*(s) = \mathbb{E}_D \left[\min_{X,B} r(X, B|s, D) + \gamma V^*(s') \right], \quad (\text{A.2})$$

where s' is the state that obtains after decisions X and B , and $\mathbb{E}_D[\cdot]$ denotes expectation taken with respect to the realization of demand. While state and action spaces are finite, it is computationally intractable to find an exact solution:²² we turn to approximate dynamic programming. In particular, we approximate the optimal value function using the Fitted Value Iteration approach ([Bertsekas \(2018\)](#), [Munos and Szepesvári \(2008\)](#)). Intuitively, in Fitted Value Iteration the classical value iteration procedure is performed only for a small subset of the states, and an estimate of the value function is computed by fitting an approximation to the values thus obtained. Following [Godfrey and Powell \(2002a,b\)](#), we choose a piecewise-linear, convex approximation; with this combination we can efficiently solve a sequence of linear and quadratic programs.

²¹We set $\gamma = 0.85$, which corresponds to an effective time horizon of approximately one month.

²²There are about $10^{2,500}$ possible states.

A.2 Fitted Value Iteration

With this approach, one first defines a set of candidate functions, and then looks for a function from this set that is closest to the fixed point of [Equation \(A.2\)](#). From [Godfrey and Powell \(2002a,b\)](#) we know that V^* must be convex in s , and it is known that a convex function can be approximated arbitrarily well with a family of affine functions.²³ Thus, we restrict attention to a set of piecewise-linear, convex functions. Each function in the set is defined as the pointwise supremum of a family of affine *basis* functions, in turn obtained from a set of basis states. Formally, let the set of basis states be $\mathcal{S} = \{s^i : i = 1, \dots, N_A\}$. Each basis state s^i is defined as follows: if N is the total number of vessels in the environment, first $\lceil \frac{3}{4}N \rceil$ are allocated uniformly at random across the $N_A \times (T_{\max} + 1)$ matrix; then we modify each entry on the i -th row as

$$s_{i,\tau}^i \leftarrow s_{i,\tau}^i \left\lceil \frac{N}{4(T_{\max} + 1)} \right\rceil.$$

Thus, s^i corresponds to a situation in which relatively more of the vessels become available in location i over time, so that we expect $V^*(s^i)$ to capture how “good” it is to have tankers in i . To each state $s^i \in \mathcal{S}$ we associate an initial value V_0^i , defined as the γ -discounted value over 147 time periods earned by a myopic central planner.²⁴ Finally, we obtain a family of affine functions by solving the following problem, where $\hat{V}_0^i \in \mathbb{R}$, $g_0^i \in \mathbb{R}^{N_A}$, and $x \cdot y$ is the usual dot product between x and y .

$$\begin{aligned} \min_{\hat{V}_0^i, g_0^i} \quad & \sum_{i=1}^{N_A} \left(\hat{V}_0^i - V_0^i \right)^2 \\ \text{s.t.} \quad & \hat{V}_0^j \geq \hat{V}_0^i + \sum_{\tau=1}^{T_{\max}+1} \gamma^{\tau-1} [g_0^i \cdot (s_{\cdot,\tau}^j - s_{\cdot,\tau}^i)] \text{ for all } i, j \end{aligned} \tag{A.3}$$

This procedure yields a set $\mathcal{B}_0^S = \left\{ (s^i, \hat{V}_0^i, g_0^i) : i = 1 \dots, N_A \right\}$. The set $\mathcal{B}_0^S$ represents basis functions because for each i we can write the affine function

$$g^i(s) = \hat{V}_0^i + \sum_{\tau=1}^{T_{\max}+1} \gamma^{\tau-1} [g_0^i \cdot (s_{\cdot,\tau} - s_{\cdot,\tau}^i)].$$

²³See, e.g., [Boyd and Vandenberghe \(2004\)](#), Chapter 3.

²⁴That is, we simulate 147 time periods and collect all the flow payoffs according to [Equation \(A.1\)](#) that a myopic planner would achieve. 147 five-day periods correspond to two years from January 1st, 2018 to January 1st, 2020.

We define our initial estimate for the value function of the DP as the pointwise supremum of these g^i 's:

$$\hat{V}_0(s) = \max_i \left\{ \hat{V}_0^i + \sum_{\tau=1}^{T_{\max}+1} \gamma^{\tau-1} [g_0^i \cdot (s_{\cdot,\tau} - s_{\cdot,\tau}^i)] \right\}. \quad (\text{A.4})$$

Note that by construction $\hat{V}_0$ is convex. The Fitted Value Iteration procedure seeks a function in the form of Equation (A.4) that approximately solves the Bellman equation in (A.2).

Procedure. The idea of the procedure is to iteratively define basis functions $\mathcal{B}_k^S$ for $k = 1, 2, \dots$ whose pointwise supremum approximates V^* better and better. Towards this end, for each k we first perform one approximate Bellman step on each basis state, yielding new values V_k^i for $i \in \mathcal{L}$. Formally,

$$V_k^i = \frac{1}{N_S} \sum_p \left[\min_{X,B} r(X, B | s^i, D^p) + \gamma \hat{V}_{k-1}(s') \right]. \quad (\text{A.5})$$

We approximate the expectation over the demand realization with a Monte Carlo method drawing N_S samples from D , independently for each state s^i . Since s' is a linear function of s^i , and $\hat{V}_0(s')$ admits a linear representation in X and B via Equation (A.4), the Bellman step can be cast as a linear integer program. We then obtain the new $\mathcal{B}_k^S$ by solving the convex fitting problem.

$$\begin{aligned} \min_{\hat{V}_k^i, g_k^i} \quad & \sum_{i=1}^{N_A} (\hat{V}_k^i - V_k^i)^2 \\ \text{s.t.} \quad & \hat{V}_k^j \geq \hat{V}_k^i + \sum_{\tau=1}^{T_{\max}+1} \gamma^{\tau-1} [g_k^i \cdot (s_{\cdot,\tau}^j - s_{\cdot,\tau}^i)] \text{ for all } i, j \end{aligned} \quad (\text{A.6})$$

The new basis functions are represented by $\mathcal{B}_k^S = \left\{ (s^i, \hat{V}_k^i, g_k^i) : i = 1 \dots, N_A \right\}$, where $\hat{V}_k^i$ and g_k^i are the optimal solutions to Equation (A.6). In turn, this procedure generates a sequence $(\hat{V}_k)_{k=1}^{\infty}$ of approximate value functions. While this sequence cannot be guaranteed to converge to a limit,²⁵ it appears from Figure 2 that the Bellman error $e_k = \|\hat{V}_{k+1} - \hat{V}_k\|_2$ quickly settles at small values, indicating that Equation (A.2) is approximately satisfied. Denote by $\hat{V}$ the approximate value function obtained with this procedure.

A.3 Comparison with Perfect Information

We use $\hat{V}$ as the approximate value function to compute which decisions the central planner would have made when facing the demand realizations observed in the data. In practice, for every $t =$

²⁵See the discussions in Gordon (1995) and Bertsekas (2018) for additional details on the reasons why Fitted Value Iteration may fail to converge.

$0, \dots, T$ we solve

$$\min_{X,B} r(X, B | s^t, D^t) + \gamma \hat{V}(s')$$

where s^t is the state representation of the situation faced by the central planner as generated by its past decisions and D^t is the demand instance in period t . We collect all optimal decisions (X^t, B^t) and then define the ballasting cost associated with them as

$$C^{DP} = \sum_{l,o,d,t} C_{l,o}^v(X_{l,o,d}^t + B_{l,o}^t).$$

As mentioned before, the central planner does not satisfy all the loads. Let $\hat{C}^{obs}$ represent the benchmark ballasting emissions, computed as follows: for every route (o, d) and every time period t we sort the loads observed in reality in ascending order of ballasting emissions, and then consider only the $n_{o,d,t}$ least costly, where $n_{o,d,t}$ is the number of loads on (o, d) in period t that were served in the simulation; $\hat{C}^{obs}$ is the sum of the ballasting emissions associated with these voyages. Then we have that

$$\frac{C^{DP}}{\hat{C}^{obs}} \approx 80\%$$

Comparing this ratio with $Share^{Trade}$, we conclude that the share of ballasting due to uncertainty is 7.5%. Because of our conservative approach to computing $\hat{C}^{obs}$, it is likely that we are underestimating this share, which is consistent with our finding in [Section 3](#) that uncertainty accounts for between 7% and 11% of ballasting costs.

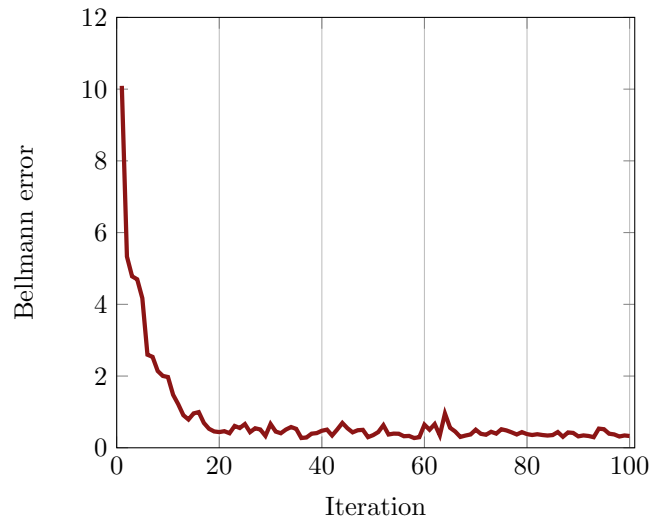


Figure 2: L_2 distance between each successive set of basis values $\hat{V}_k^i$ for the basis states of the Fitted Value Iteration procedure.

B Econometric details

The purpose of this appendix is to provide formal definitions and more details on the econometric approaches that we used in [Section 3.4](#) and [Sections 4.2](#) and [4.3](#). We discuss each instrumental variable we construct, how it is used, and argue for its validity. Finally, we present the estimates for the specifications that we omitted from the main body.

B.1 Instrumental Variables

There are two endogenous covariates that we need instrumental variables for: market concentration in [Section 3.4](#) and pool size in [Sections 4.2](#) and [4.3](#). Market concentration is defined at aggregate level, while pool size is defined at operator level; our choice of IVs follows the respective level of granularity. In all cases, the objective is to isolate exogenous variation in fleet size or market concentration that is not driven by contemporaneous local demand conditions or operator-specific shocks affecting outcomes directly.

B.1.1 IVs for market concentration

We measure market concentration using the Herfindahl–Hirschman Index (HHI), defined as the sum of the squared market shares of each operator. Formally, we define as *market* each pair of load area l and quarter t . For each market (l, t) and vessel class v , let $\mathcal{P}_{l,t,v}$ be the set of operators who served at least one load originating from load area l in period t . For each $i \in \mathcal{P}_{l,t,v}$, define the market share as

$$s_{i,l,t,v} = \frac{V_{i,l,t,v}}{\sum_{j \in \mathcal{P}_{l,t,v}} V_{j,l,t,v}},$$

where $V_{i,l,t,v}$ denotes the number of voyages performed by operator i in market (l, t) within vessel class v . Thus, market shares are computed using the number of realized voyages, which we treat as a proxy for operator activity in a given market. The HHI is then defined as

$$HHI_{l,v,t} = \sum_{i \in \mathcal{P}_{l,t,v}} s_{i,l,t,v}^2.$$

The risk that $HHI_{l,v,t}$ is endogenous to average ballast measures stems from potential reverse causality, as we argue in [Section 3.4](#): after observing that a certain market (l, t) is experiencing lower than usual ballast times (e.g., a positive demand shock), commercial operators may further concentrate their operations in that market.

Lagged HHI. Our first choice of IV for market concentration is simply the lagged HHI of the same market. Formally,

$$HHI_{l,v,t}^{\text{lag}} = HHI_{l,v,t-1}.$$

Because our specification already include (l, t) and vessel class fixed effects, we only need to address endogeneity arising from shocks that make ballasting to a load area *temporarily* less efficient. Given that ballasting trips are long, quarter-to-quarter HHI changes are driven by slow-moving entry and exit decisions that naturally precede contemporaneous ballast outcomes.

Bartik-style IVs. As a form of robustness, we verify that our estimates are robust to using a different type of instrumental variable. Since we are dealing with *market-level* outcomes, we use Bartik-style instruments (Goldsmith-Pinkham et al., 2020). Bartik IVs (also called “shift-share” IVs) combine predetermined market shares with aggregate operator-level shocks, thereby generating predicted variation in market structure that is plausibly exogenous to contemporaneous local conditions. Formally, for each operator i we identify a market share $s_{i,l,0,v}$ in a baseline period.²⁶ Let also $V_{i,t,v}$ denote the total number of voyages served by i in period t and vessel class v (across all load areas). We then construct “predicted” market shares as

$$\hat{s}_{i,l,t,v} = \frac{s_{i,l,0,v} \cdot V_{i,t,v}}{\sum_j s_{j,l,0,v} \cdot V_{j,t,v}},$$

where summation is over *all* observed commercial operators of a certain vessel class. With predicted market shares we define the Bartik-HHI as

$$Z_{l,t,v}^{\text{Bartik}} = \sum_i \hat{s}_{i,l,t,v}^2.$$

The identifying variation comes from changes in operator activity $V_{i,t,v}$, which is aggregated at the global level and thus plausibly orthogonal to idiosyncratic shocks affecting a market (l, t) . The key identifying assumption is that aggregate changes in operator activity be driven by factors not specific to any single market. Baseline shares are predetermined and therefore do not respond to contemporaneous shocks.

This construction is particularly well suited to the maritime shipping setting. First, operators typically maintain relatively stable geographic footprints over time, reflecting long-standing relationships with brokers, charterers, and trading routes. As a result, baseline shares $s_{i,l,0,v}$ capture persistent exposure of each operator to specific markets. Second, fluctuations in $V_{i,t,v}$ are largely

²⁶In general, we use as baseline share of operator i the first period t in which we observe voyages served by i from load area l using vessel class v .

driven by global factors—such as fleet expansion, entry and exit, financing conditions, or shifts in aggregate demand for shipping capacity—that affect operators across all markets simultaneously. The interaction of these aggregate shocks with predetermined exposure shares generates variation in predicted market concentration that is not driven by local conditions in a given load area.

Importantly, because the Bartik-HHI is constructed using operator-level activity aggregated across all markets, it is not mechanically affected by shocks specific to a particular (l, t) market, except through the predetermined shares. In this sense, the instrument isolates changes in concentration arising from the reallocation of globally determined operator capacity across markets, rather than from local demand fluctuations or equilibrium responses within a given market. This makes it a natural instrument for market concentration in our context, where operators simultaneously participate in multiple geographically dispersed markets and where global capacity decisions play a central role in shaping local market structure.

To reduce reliance on a fixed baseline period and to allow for gradual changes in market structure, we also construct a rolling version of the Bartik instrument. In this case, baseline shares are taken from a lagged period $t - \ell$, where ℓ corresponds to four quarters:

$$\tilde{s}_{i,l,t,v} = \frac{s_{i,l,t-\ell,v} \cdot V_{i,t,v}}{\sum_j s_{j,l,t-\ell,v} \cdot V_{j,t,v}},$$

This specification allows the allocation of aggregate shocks to evolve over time while maintaining the shift-share structure that underpins identification. The corresponding IV for market concentration is then

$$Z_{l,t,v}^{\text{Bartik-R}} = \sum_i \tilde{s}_{i,l,t,v}^2.$$

First-stage diagnostics. The first-stage results reported in [Table 8](#) indicate that all instruments are positively and significantly correlated with market concentration. The specification using lagged concentration exhibits a particularly strong first stage, with a large and precisely estimated coefficient and a Wald statistic well above conventional thresholds. This reflects the persistence of market structure over time and confirms that lagged HHI is a powerful predictor of current concentration. The Bartik instrument based on baseline shares and aggregate operator activity also displays a strong first stage, with a Wald statistic exceeding standard weak-instrument thresholds. The magnitude of the coefficient suggests that predicted concentration closely tracks realized concentration, consistent with stable operator presence across markets.

The rolling Bartik specification yields a smaller coefficient and a lower Wald statistic, indicating a weaker first stage. This attenuation is expected, as using lagged shares reduces the mechanical relationship between predicted and realized concentration while allowing exposure patterns to

Dependent Variable:	Log (HHI)		
<i>Instrumental Variables</i>			
Log (Lagged HHI)	0.5290*** (0.0527)		
Log (Bartik HHI)		0.7142*** (0.2028)	
Log (Rolling Bartik HHI)			0.1974*** (0.0726)
<i>Fixed-effects</i>			
Load Area	Yes	Yes	Yes
Quarter	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	2,143	2,159	1,641
R ²	0.93446	0.91352	0.92271
Within R ²	0.29674	0.12129	0.05609
Wald (1st stage)	100.85	12.397	7.3872
<i>Clustered (Load Area) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Table 8: First-stage regressions of HHI on the lagged HHI and Bartik-style IVs.

evolve over time. Although this specification is appealing from an identification standpoint, the weaker first-stage relationship suggests greater sensitivity to weak-instrument concerns. Across all specifications, the inclusion of load-area and time fixed effects implies that identification relies on within-market variation over time, net of persistent differences across locations and common shocks. Taken together, the results suggest that the lagged HHI and baseline Bartik instruments provide strong sources of variation, while the rolling Bartik specification serves as a more conservative robustness check.

Robustness check. We present here the estimates from using the same specifications as [Section 3.4](#) in a regression that employs the Bartik-style instrumental variables. Comparing [Table 9](#) with [Table 5](#) we see that the estimates based on Bartik-style instruments are broadly consistent with those obtained using lagged HHI as an instrument, both in sign and economic magnitude. Across all outcomes, higher market concentration is associated with lower ballast distance, emissions, and time, and the IV estimates are generally larger in absolute value than their OLS counterparts. The

Bartik specifications yield coefficients that are comparable to, and in some cases larger than, those obtained with the lagged HHI instrument, despite relying on a different source of identifying variation. While the rolling Bartik instrument produces somewhat larger point estimates, it is associated with a weaker first stage. Overall, the similarity of the results across these alternative instruments suggests that the estimated relationship between market concentration and operational efficiency is not driven by the specific choice of instrument, but rather reflects a robust empirical pattern.

	Ballast Distance (nm)			Ballast Emissions (CO2)			Ballast Time (days)		
	OLS	Bartik	Rolling Bartik	OLS	Bartik	Rolling Bartik	OLS	Bartik	Rolling Bartik
<i>Variables</i>									
Log (HHI)	-0.2581*** (0.0543)	-0.2911** (0.1217)	-0.4111*** (0.1495)	-0.2428*** (0.0461)	-0.2878*** (0.1050)	-0.3992*** (0.1457)	-0.2026*** (0.0458)	-0.1994** (0.0963)	-0.2664*** (0.0929)
<i>Fixed-effects</i>									
Load Area	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>									
Observations	2,297	2,159	1,641	2,297	2,159	1,641	2,297	2,159	1,641
R ²	0.90674	0.99310	0.99324	0.94668	0.99502	0.99458	0.93063	0.99571	0.99609
Wald (1st stage F-stat)	—	12.397	7.3872	—	12.397	7.3872	—	12.397	7.3872
<i>Clustered (AreaID) standard-errors in parentheses</i>									
<i>Signif. Codes: ***, 0.01, **, 0.05, *, 0.1</i>									

Table 9: Market-level regressions of ballast efficiency measures on market concentration (HHI) with Bartik-style IVs.

Notes: Standard errors clustered at the Load Area $\times$ Vessel Class level. Regressions weighted by number of trips. From the total sample of $28 \times 6 \times 16 = 2,688$ market-quarter pairs, we restrict attention to those with at least 10 departing loads, yielding 2,297 market-quarter observations. For IV regressions, sample size is smaller due to some markets not having a well-defined IV.

B.1.2 IVs for pool size

We recall from [Section 4.2](#) that we define the size of the pool of operator i during time period t as the (time-)weighted number of vessels that the operator controls, and we argued that there exist substantial concern of endogeneity of this measure with efficiency measures at operator-time period level: the most worrisome are unobservable improvements of the operator’s management and reverse causality. Addressing these endogeneity risks requires a source of exogenous, yet prolonged, variation in pool sizes. Exploiting natural experiments is challenging because of the long travel times in shipping: the 2021 Suez Canal blockage, for example, affected fleet availability for only six days, whereas the average voyage lasts 25 days. To have a detectable impact on pool-wide outcomes, exogenous supply shocks would need to persist for several weeks or months. For these reasons, we opt to use Hausman-style instrumental variables: the underlying idea of these IVs is that pool size changes are driven by both operator-specific idiosyncratic factors and *market-wide* unobservable factors that are orthogonal to those operator-level shocks ([Hausman, 1996](#); [Nevo, 2001](#)). This assumption appears appropriate in the maritime shipping context: because building

tankers takes several years, additions to an operator’s pool by direct purchase are planned long in advance, and therefore shipbuilders’ order books are influenced primarily by long-term forecasts about oil transportation, which are available to all players in the market.

We now present two definitions of Hausman-style IVs. Let $F_{i,v,t}$ denote the fleet size of operator i in vessel class v during period t : at a high level, a Hausman instrument for $F_{i,v,t}$ is a (appropriately defined) average of the pool size of operators that are “comparable” to i at time t , and that don’t affect i ’s operations directly. Comparability is necessary to ensure that the average captures the “common” underlying factor affecting i ’s pool size, while the latter condition is necessary to satisfy the exclusion restriction for the validity of IVs. To define these “peer” groups, we partition operators along two dimensions. Specifically, for each vessel class v and period t :

- (i) We partition operators into quartiles based on their fleet size: let $Q(i, v, t)$ denote the set of operators in the same fleet-size quartile as operator i in vessel class v at time $t - 1$.
- (ii) We partition operators according to their “market focus”: operator i is classified as *focused* if there exists a load area $l(i, t)$ such that more than half of its voyages originate from $l(i, t)$ in period t , and *global* otherwise. Let $G(i, v, t) \subseteq Q(i, v, t)$ denote the subset of operators in the same quartile and with the same market-focus classification as i .

Same vessel-class Hausman IV (*Same VC*). Our first IV for operator i in vessel class v at time t uses variation in the pool size of operators that are comparable to i and use the same vessel class. Formally, we define the same-class Hausman instrument as:

$$Z_{i,v,t}^{\text{SameVC}} = \frac{\sum_{j \in Q(i,v,t)} F_{j,v,t} - \sum_{j \in G(i,v,t)} F_{j,v,t}}{|Q(i, v, t)| - |G(i, v, t)|}.$$

In other words, we instrument $F_{i,v,t}$ with the average fleet size of operators whose market focus is *different* from that of operator i . This ensures a certain degree of separation between operator i and those in $Q(i, v, t) \setminus G(i, v, t)$, because these operators are less likely to directly compete on the same loads as operator i . At the same time, $F_{i,v,t}$ and $Z_{i,v,t}^{\text{SameVC}}$ are still exposed to the same shocks that affect specific vessel classes, such as canal closures/openings or new environmental regulations in ports that can be visited only by vessels of given size. As a measure of robustness, we consider also an instrumental variable that uses variation from other vessel classes.

Cross vessel-class Hausman IV (*Cross VC*). Let $\mathcal{V}$ be the set of all vessel classes and further let

$$Q(i, t) = \bigcup_{v \in \mathcal{V}} Q(i, v, t)$$

denote the set of operators in the same size quartile as i when pooling across all vessel classes, and $G(i, t) \subseteq Q(i, t)$ similarly denote the subset of these operators with the same market focus as i . We define the cross-class Hausman instrument as:

$$Z_{i,v,t}^{\text{CrossVC}} = \frac{\left(\sum_{j \in Q(i,t)} F_{j,t} - \sum_{j \in Q(i,v,t)} F_{j,v,t} \right) - \left(\sum_{j \in G(i,t)} F_{j,t} - \sum_{j \in G(i,v,t)} F_{j,v,t} \right)}{N_{i,v,t}},$$

where put $F_{j,t} = \sum_{v \in \mathcal{V}} F_{j,v,t}$ and

$$N_{i,v,t} = | [Q(i, t) \setminus Q(i, v, t)] \setminus [G(i, t) \setminus G(i, v, t)] |$$

is the number of operators in the peer group. In other words, the peer group is composed of all those operators in the same size quartile as i , excluding those in the same vessel class and those with similar market focus. Ultimately, the aim of this construction is to further reduce the scope for direct competitive or equilibrium interactions, which are more likely to arise within the same vessel class.

First-stage diagnostics. The first-stage results reported in [Table 10](#) show that both Hausman-style instruments are strong predictors of pool size. In both specifications, the coefficient on the instrument is positive, precisely estimated, and statistically significant. The magnitudes are similar across the two constructions, indicating that increases in peer fleet size translate proportionally into increases in the operator’s own pool size. The corresponding Wald statistics are very large, exceeding conventional thresholds for weak instruments and indicating a strong first stage. These results confirm that variation in peer fleet size—whether constructed within or across vessel classes—is highly informative about operator-level fleet decisions.

Comparing the two specifications, the cross VC IV performs similarly to the same VC Hausman IV in terms of both magnitude and statistical strength. This is notable because the cross-VC construction is designed to mitigate concerns about within-market equilibrium effects and common shocks affecting operators in the same vessel class. The fact that it delivers a first stage of comparable strength suggests that a substantial portion of the relevant variation in fleet size is driven by broader industry dynamics rather than vessel-class-specific factors. Taken together, these results provide strong support for the relevance of the Hausman-style instruments and indicate that both specifications offer credible sources of variation, with the cross-VC instrument providing a particularly appealing balance between strength and exogeneity.

Dependent Variable:		log (Pool Size)
<i>Instrumental Variables</i>		
log(IV Same VC)	0.3974*** (0.0225)	
log(IV Cross VC)		0.4014*** (0.0233)
Granularity	Month	Month
<i>Fixed-effects</i>		
Operator $\times$ Vessel Class	Yes	Yes
Month	Yes	Yes
<i>Fit statistics</i>		
Observations	24,455	24,455
R ²	0.91665	0.91731
Within R ²	0.19786	0.20422
Wald (1st stage)	310.76	297.26
<i>Clustered (Commercial Operator) standard-errors in parentheses</i>		
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>		

Table 10: First-stage regressions of pool size on the two Hausman-style IVs.

B.2 Additional specifications

Dependent Variable: Model:	Distance-utilization		
	OLS	IV Same VC	IV Cross VC
<i>Variables</i>			
$\log(PoolSize)$	0.0137*** (0.0036)	0.0095 (0.0062)	0.0099* (0.0060)
Granularity	Month	Month	Month
<i>Fixed-effects</i>			
Operator $\times$ Vessel Class	Yes	Yes	Yes
Month	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	24,455	24,455	24,455
R ²	0.23187	0.23088	0.23089
Wald (1st stage F-stat)	N/A	310.76	297.26
<i>Clustered (Commercial Operator) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Table 11: Robustness: Regression of Equation (6) estimated using a *distance-based* definition of utilization. Distance-utilization for operator i during period t is defined as $Utilization_{i,t}^D = \frac{\sum_n D_{n,i,t}^L}{\sum_n D_{n,i,t}^V}$, where $D_{n,i,t}^V$ and $D_{n,i,t}^L$ are total voyage and laden distance respectively of voyage n . Estimates are consistent in sign and magnitude with Table 6, although standard errors for the IV estimates are larger. This is generally due to traveled distances exhibiting less variation across voyages compared to travel times.

Dependent Variable: Model:	OLS	Ballast Delay IV Same VC	IV Cross VC
<i>Independent Variable</i>			
$\log(PoolSize)$	-0.1689* (0.0899)	-0.4053* (0.2227)	-0.3610* (0.2085)
Granularity	Quarter	Quarter	Quarter
<i>Fixed-effects</i>			
Commercial Operator $\times$ Vessel Class	Yes	Yes	Yes
Period	Yes	Yes	Yes
Starting-Load port pair	Yes	Yes	Yes
Geographic granularity	Narrow Area	Narrow Area	Narrow Area
Time Granularity	Month	Month	Month
<i>Fit statistics</i>			
Observations	175,257	175,257	175,257
R ²	0.11997	0.11997	0.11996
Wald (1st stage F-stat)	N/A	127.05	139.76
<i>Clustered (Commercial Operator) standard-errors in parentheses</i>			
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>			

Table 12: Robustness: Regression of Equation (7) estimated at Narrow Area level. There are 51 narrow areas compared to 28 areas, so sample size increases. Estimates are broadly consistent with Table 7, although standard errors are larger.

Notes: Because imposing fixed effects for each port pair exponentially increases the number of variables to estimate, with resulting loss of statistical power, we estimate origin-destination fixed effects at narrow area level.

Panel A: Ballast Distance (nm)				
	OLS	Lagged HHI IV	Bartik IV	Rolling Bartik IV
<i>Independent Variable</i>				
Log(HHI)	-0.2530*** (0.0471)	-0.3558*** (0.0733)	-0.3726*** (0.1254)	-0.5574*** (0.1667)
<i>Fixed-effects</i>				
Narrow Area	Yes	Yes	Yes	Yes
Quarter	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	3,683	3,406	3,378	2,556
R ²	0.89294	0.98782	0.99041	0.99037
Within R ²	0.05267	0.88933	0.91189	0.90356
Wald (1st stage F-stat)		160.74	15.258	10.627
Panel B: Ballast Emissions (CO₂)				
	OLS	Lagged HHI IV	Bartik IV	Rolling Bartik IV
<i>Independent Variable</i>				
Log(HHI)	-0.2390*** (0.0403)	-0.3357*** (0.0638)	-0.3546*** (0.1074)	-0.5336*** (0.1483)
<i>Fixed-effects</i>				
Narrow Area	Yes	Yes	Yes	Yes
Quarter	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	3,683	3,406	3,378	2,556
R ²	0.93673	0.99151	0.99313	0.99291
Within R ²	0.05861	0.86973	0.89367	0.87784
Wald (1st stage F-stat)		160.74	15.258	10.627
Panel C: Ballast Time (days)				
	OLS	Lagged HHI IV	Bartik IV	Rolling Bartik IV
<i>Independent Variable</i>				
Log(HHI)	-0.1940*** (0.0355)	-0.2639*** (0.0526)	-0.2472** (0.1024)	-0.3461*** (0.1016)
<i>Fixed-effects</i>				
Narrow Area	Yes	Yes	Yes	Yes
Quarter	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	3,683	3,406	3,378	2,556
R ²	0.91892	0.99051	0.99269	0.99329
Within R ²	0.06421	0.88864	0.91190	0.91049
Wald (1st stage F-stat)		160.74	15.258	10.627
<i>Clustered (Narrow Load Area) standard-errors in parentheses</i>				
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

Table 13: Robustness: Regression of Equation (5) estimated at Narrow Area level. There are 51 narrow areas compared to 28 areas, so sample size increases. Estimates are broadly consistent with Tables 5 and 9.

C Additional tables and figures

C.1 Tables

Vessel Class	Ballast distance (nm)	Ballast portion (%)	CO ₂ emissions (tons)
MR1	1,054	40.2	218.8
MR2	1,411	38.6	293.8
Panamax	1,571	40.9	377.5
Aframax	1,424	42.1	443.8
Suezmax	2,611	43.7	933.1
VLCC	5,718	45.7	2943.71

Table 14: Average ballasting distance, average portion of ballasting on total voyage length and average emissions due to ballasting, broken down by vessel class.

Vessel class	Trade imbalances (%)	Uncertainty (%)	Fragmentation (%)
MR1	76.76	12.17	11.07
MR2	74.10	12.11	13.79
Panamax	73.41	15.43	11.16
Aframax	72.16	16.55	11.29
Suezmax	77.81	12.22	9.97
VLCC	87.06	7.79	5.15

Table 15: Share of ballasting attributed to trade imbalances, uncertainty and market fragmentation, using as objective function in [Section 3](#) total fuel consumption associated with ballasting travels.

Geographical Area	Avg. arrivals	Avg. departures
Red Sea	13.9	12.5
West Africa	21	26.9
Pacific Islands	2.1	0.2
Russian Pacific	0.1	7.6
Caribs	19.6	21.8
Baltic	11.9	31.8
South East Asia	60.5	51.8
US Gulf & Mainland	25.2	60.3
Korea / Japan	34.1	21.9
Black Sea / Sea Of Marmara	8	30.3
East Coast South America	21.3	20.5
West Coast Mexico	4.8	2.1
West Coast North America	13.1	8.2
West Coast South America	10.9	7.8
Australia / New Zealand	12.1	5.3
India / Pakistan	39.7	19.9
East Coast Canada	5.7	4.3
East Coast Central America	3.7	0.6
US Atlantic Coast	19.9	1.6
South East Africa	9.7	1.4
North Sea	1.2	5.4
UK Continent	59.2	31.3
West Coast Central America	4.2	2.3
East Coast Mexico	9.5	6.8
China / Taiwan	62.1	23.5
Mediterranean	68.1	56
Arabian Gulf	21.4	97.6
Arctic Ocean & Barents Sea	4	7.2

Table 16: Average number of ships arriving laden and departing laden from each geographical area. The table summarizes loads for all vessel classes, and the figures are obtained by looking at the average number of loads arriving/departing in windows of 5 days.

Commercial Operator	Pool size	Share of vessels(%)	Share of tonnage(%)
Teekay Corp	40	3.9	4.2
Sovcomflot	35	3.4	3.7
AET	33	3.2	3.4
Scorpio Commercial Management	27	2.6	2.8
Minerva Marine	23	2.2	2.4
ST Shipping & Transport	22	2.1	2.2
Cardiff Marine	21	2	2.3
Thenamaris	20	2	2.1
Shell	20	2	2.1
Navig8 group	20	2	2.1
Heidmar	18	1.8	1.9
Vitol	16	1.6	1.7
Trafigura	16	1.6	1.7
Equinor	15	1.5	1.7
Penfield Marine	15	1.5	1.6
Maersk	12	1.2	1.2
Signal Maritime	12	1.2	1.2
Zodiac Maritime	12	1.2	1.3
ExxonMobil	12	1.2	1.3
Frontline	12	1.2	1.2

Table 17: Average pool size for the 20 largest operators in the Aframax segment. The share of vessels indicates the percentage of active Aframax ships observed in the dataset controlled by each commercial operator, and the share of tonnage indicates what share of total capacity of the segment is controlled by each commercial operator.

C.2 Figures

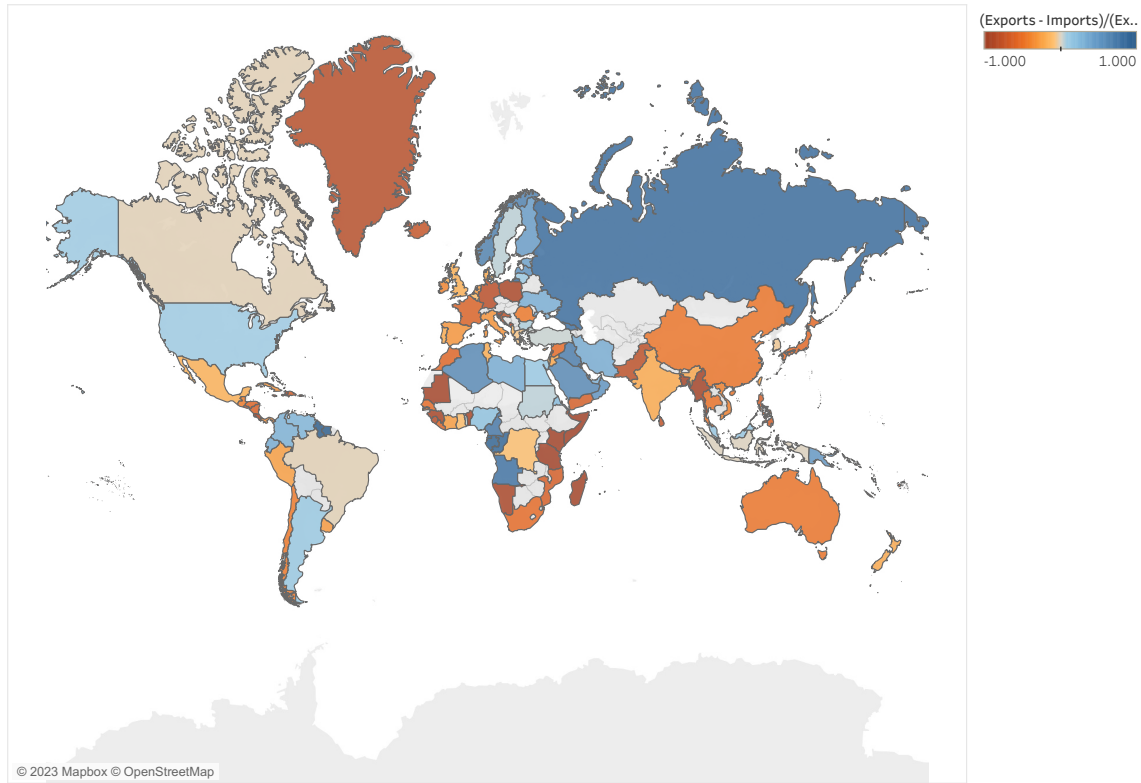


Figure 3: A map showing the demand imbalances across the network. The color of each country reports the difference between the number of discharge events and the number of loading events. A positive value indicates a country with more exports than imports, while a negative value points to countries with higher imports than exports. China is the largest importer, while Russia and Saudi Arabia are the largest exporters.

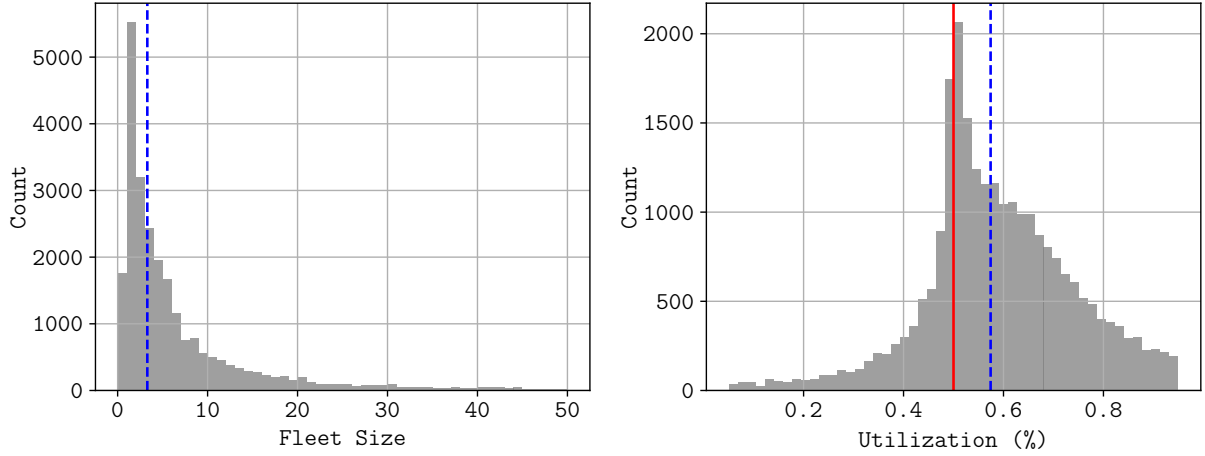


Figure 4: The left histogram shows the variation in fleet size at a quarterly level, i.e., for each quarter and commercial operator we measure their average fleet size and generate the histogram using these values. The plot illustrates the market structure, emphasizing its fragmentation (most of the observations are on the left side). The blue dotted line reports the median of the distribution, which is equal to 3.3. Similarly, the right panel shows the utilization (laden miles/total miles) of each fleet at a quarterly level. The red solid line is at the 0.5 level, where all trips would be of the style *out and back*, and the dotted blue line marks the median of the distribution (56.8%).

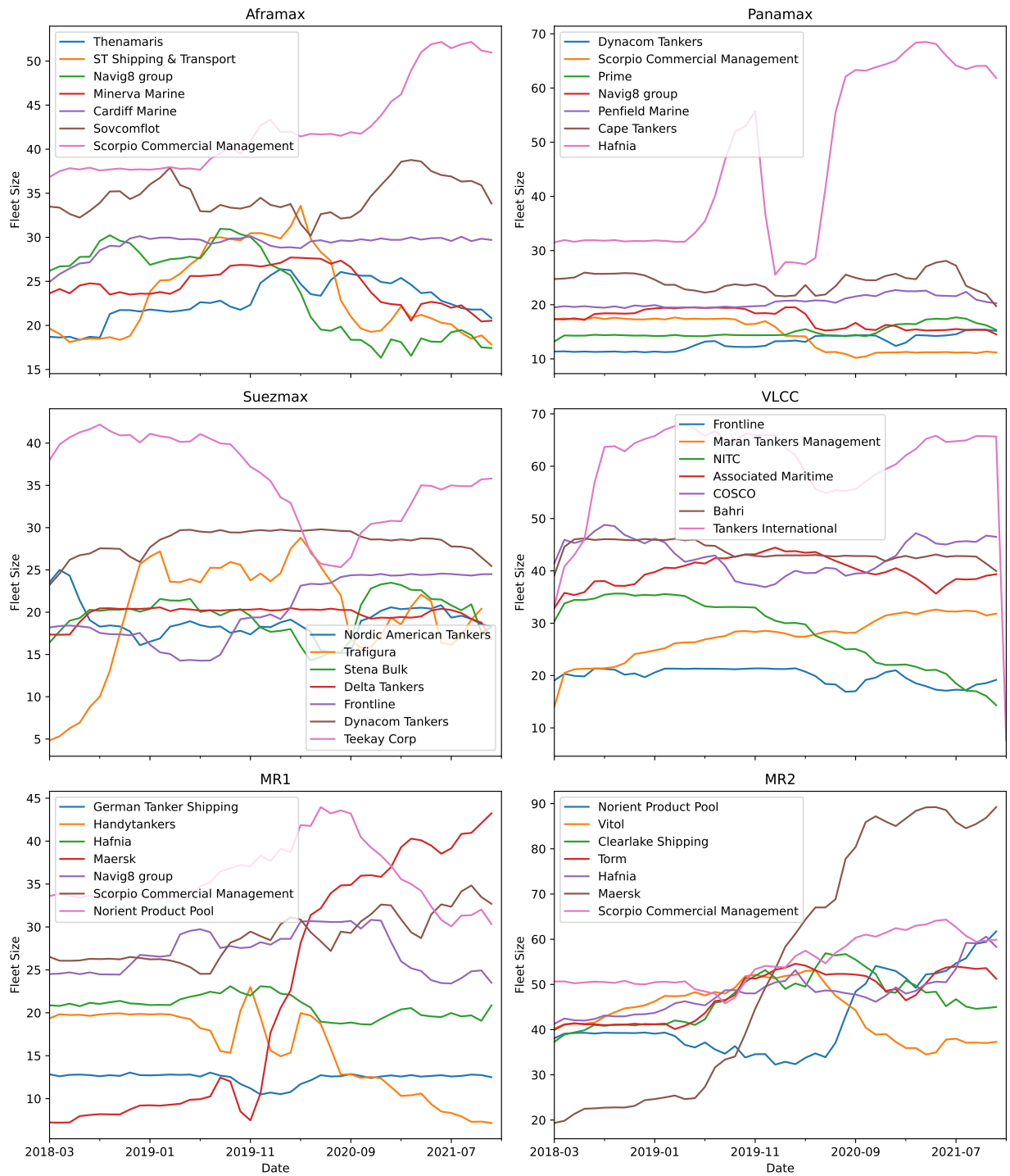


Figure 5: The figure shows the variation across time of the average fleet size of the seven largest commercial operators in each sub-market at a monthly level.

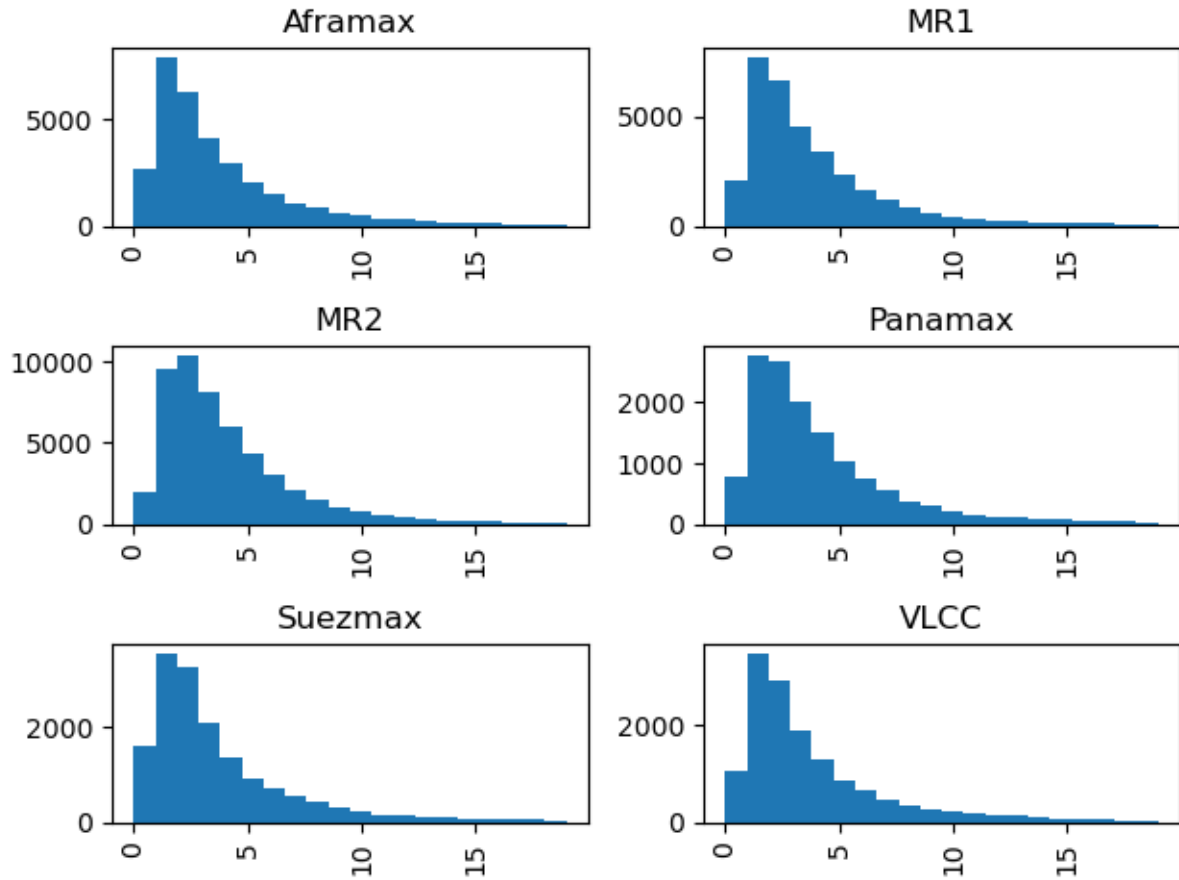


Figure 6: Distribution of the number of days between arrival of an empty vessel at port and its departure laden (i.e., loading time)

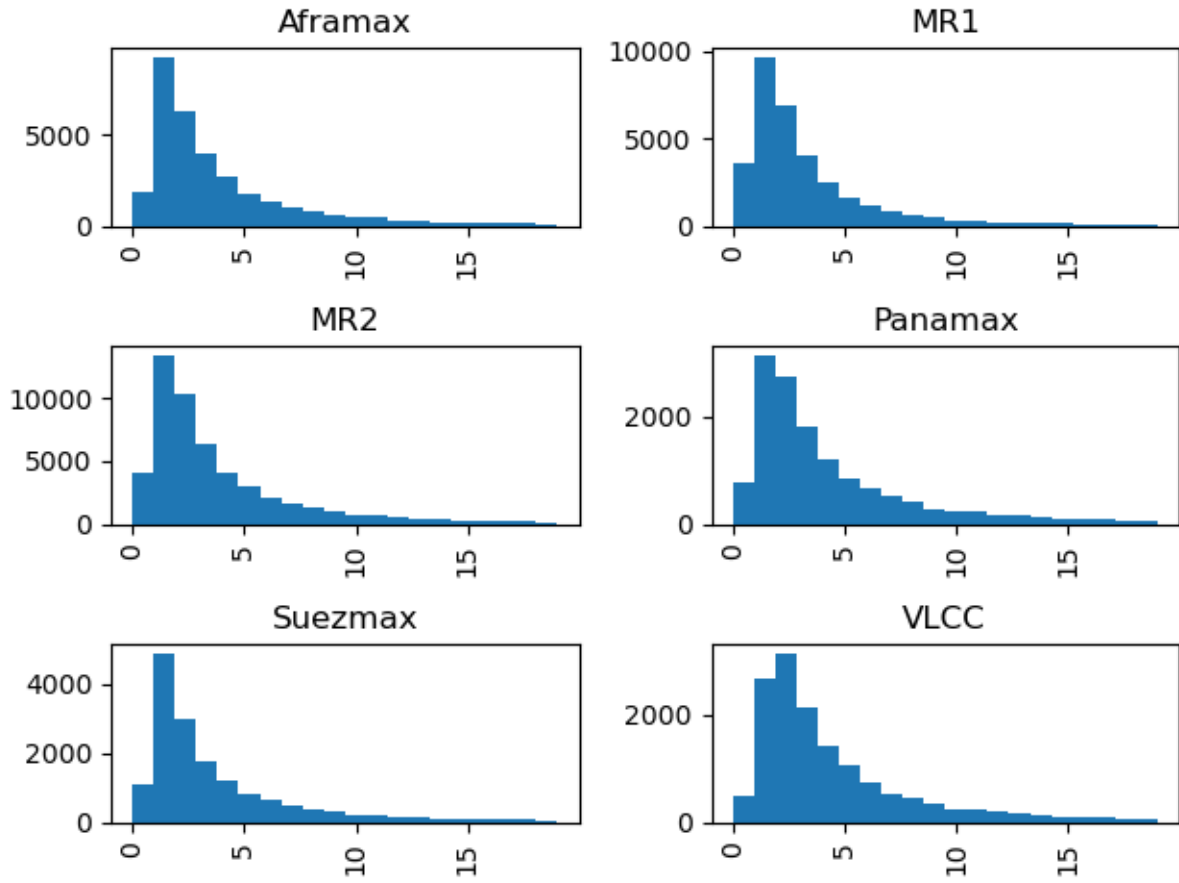


Figure 7: Distribution of the number of days between arrival of a laden vessel at port and its departure empty (i.e., unloading time)